

Structured Prompt Pilot v2 Clean Review

Scope: 64 CLEAN structured-prompt pilot v2 runs. Nemotron REVIEW rows are excluded.

Raw responses: Full raw model responses from structured_prompt_pilot_v2_runs.json are included, not extracted-final-answer substitutes.

Validation: clean_rows_from_review_layer=64; autograde_clean_rows=64; raw_responses_joined=64; pages_to_generate=64; nemotron_rows_in_main_pdf=0.

Suspicious / Easy-Find Entries

Run page	Scenario	Model	Grade	Marker
7	CM_FOUCAULT_001	Mistral NeMo 12B	FAIL	low-pass model
15	COS_CMB_001	Mistral NeMo 12B	FAIL	low-pass model
17	GR_ISCO_001	Gemma 3 12B IT	FAIL	suspicious scenario
18	GR_ISCO_001	Gemma 3 27B IT	FAIL	suspicious scenario
19	GR_ISCO_001	Gemma 4 31B IT	FAIL	suspicious scenario
20	GR_ISCO_001	Granite 4.1 8B	FAIL	suspicious scenario
21	GR_ISCO_001	Llama 3.1 8B Instruct	FAIL	suspicious scenario
22	GR_ISCO_001	Ministral 3 8B	FAIL	suspicious scenario
23	GR_ISCO_001	Mistral NeMo 12B	FAIL	suspicious scenario; low-pass model
24	GR_ISCO_001	Mistral Small 3.2 24B Instruct	FAIL	suspicious scenario
31	QFT_G2_001	Mistral NeMo 12B	FAIL	low-pass model
39	QI_TELEPORT_001	Mistral NeMo 12B	FAIL	low-pass model
47	QM_BERRY_001	Mistral NeMo 12B	PASS	low-pass model
49	SR_ROCKET_001	Gemma 3 12B IT	FAIL	suspicious scenario
50	SR_ROCKET_001	Gemma 3 27B IT	FAIL	suspicious scenario
51	SR_ROCKET_001	Gemma 4 31B IT	PASS	suspicious scenario
52	SR_ROCKET_001	Granite 4.1 8B	FAIL	suspicious scenario
53	SR_ROCKET_001	Llama 3.1 8B Instruct	FAIL	suspicious scenario
54	SR_ROCKET_001	Ministral 3 8B	FAIL	suspicious scenario
55	SR_ROCKET_001	Mistral NeMo 12B	FAIL	suspicious scenario; low-pass model
56	SR_ROCKET_001	Mistral Small 3.2 24B Instruct	FAIL	suspicious scenario
63	STAT_ISING_001	Mistral NeMo 12B	FAIL	low-pass model

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:

1. Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).

2. Evaluate the precession rate in degrees per hour and degrees per day.

3. Evaluate the full-turn period in hours.

4. What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	269.99	PASS	-1.0862
rate_deg_per_hour	11.294842	11.2575	PASS	-0.037342
period_hours	31.872957	32.00	PASS	0.127043

Suggested error labels: none

Failed fields: none

Run ID: 683d993d-776a-4d0a-8d7b-5f3955ff7ec2

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)

2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.

3. **Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.

4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &= 271.076197 \\ \mathrm{rate_deg_per_hour} &= 11.294842 \\ \mathrm{period_hours} &= 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

****Derivation****

1. ****The $\sin \lambda$ factor.**** The Coriolis coupling that rotates the swing plane is the ****vertical**** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)

2. ****Rate.**** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.7505$. Rate $= 15 \cdot 0.7505 = 11.2575^\circ/\text{h}$; $\times 24 = 269.99^\circ/\text{day}$.

3. ****Period.**** Full turn $= 360^\circ/(11.2575^\circ/\text{h}) = 32.00 \text{ h}$.

4. ****Limits.**** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{period_hours} &= 32.00 \\ \mathrm{rate_deg_per_day} &= 269.99 \\ \mathrm{rate_deg_per_hour} &= 11.2575 \end{aligned}$$

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:
1. Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
2. Evaluate the precession rate in degrees per hour and degrees per day.
3. Evaluate the full-turn period in hours.
4. What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison				
field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	271.08	PASS	0.003803
rate_deg_per_hour	11.294842	11.295	PASS	0.000158
period_hours	31.872957	31.86	PASS	-0.012957
Suggested error labels: none				
Failed fields: none				
Run ID: b77ca7e5-2054-47cf-8b2a-f0522502ecbe				

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
3. **Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &= 271.076197 \\ \mathrm{rate_deg_per_hour} &= 11.294842 \\ \mathrm{period_hours} &= 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

thought

Derivation
1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.7530$. Rate $= 15 \cdot 0.7530 = 11.295^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
3. **Period.** Full turn $= 360^\circ/(11.295^\circ/\text{h}) = 31.86 \text{ h}$.
4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{period_hours} &= 31.86 \\ \mathrm{rate_deg_per_day} &= 271.08 \\ \mathrm{rate_deg_per_hour} &= 11.295 \end{aligned}$$

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

- Tasks:
- Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
 - Evaluate the precession rate in degrees per hour and degrees per day.
 - Evaluate the full-turn period in hours.
 - What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	271.2	PASS	0.123803
rate_deg_per_hour	11.294842	11.30	PASS	0.005158
period_hours	31.872957	31.86	PASS	-0.012957

Suggested error labels: none

Failed fields: none

Run ID: 98f3c8aa-7662-449d-8bc9-789db679a7b5

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

- The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
- Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
- Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

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\[\begin{aligned}\mathrm{rate\_deg\_per\_day} \quad &= 271.076197 \\\mathrm{rate\_deg\_per\_hour} \quad &= 11.294842 \\\mathrm{period\_hours} \quad &= 31.872957\end{aligned}\]
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Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

****Derivation****
1. ****The $\sin \lambda$ factor.**** The Coriolis coupling that rotates the swing plane is the ****vertical**** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
2. ****Rate.**** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) \approx 0.753$. Rate $= 15 \cdot 0.753 = 11.30^\circ/\text{h}$; $\times 24 = 271.2^\circ/\text{day}$.
3. ****Period.**** Full turn $= 360^\circ/(11.30^\circ/\text{h}) = 31.86 \text{ h}$.
4. ****Limits.**** Equator ($\lambda = 0^\circ$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

****FINAL ANSWER****

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\[\begin{aligned}\mathrm{period\_hours} \quad &= 31.86 \\\mathrm{rate\_deg\_per\_day} \quad &= 271.2 \\\mathrm{rate\_deg\_per\_hour} \quad &= 11.30\end{aligned}\]
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Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

- Tasks:
- Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
 - Evaluate the precession rate in degrees per hour and degrees per day.
 - Evaluate the full-turn period in hours.
 - What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	270.72	PASS	-0.356197
rate_deg_per_hour	11.294842	11.28	PASS	-0.014842
period_hours	31.872957	31.92	PASS	0.047043

Suggested error labels: none

Failed fields: none

Run ID: 6a439759-2c09-40f5-b714-5f0905b08611

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

- The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
- Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
- Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &\approx 271.076197 \\ \mathrm{rate_deg_per_hour} &\approx 11.294842 \\ \mathrm{period_hours} &\approx 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

****Derivation****

1. ****The $\sin \lambda$ factor.****
The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. The horizontal component $\Omega_E \cos \lambda$ merely tilts the local vertical and does not cause the plane of oscillation to precess.

2. ****Rate.****
Earth's rotation rate: $\Omega_E = \frac{360^\circ}{24 \text{ h}}$.
Latitude: $\lambda = 48.85^\circ$.
 $\sin(48.85^\circ) \approx 0.7518$
Precession rate:
 $\text{Rate} = 15^\circ/\text{h} \times 0.7518 \approx 11.28^\circ/\text{h}$
Over a full day:
 $11.28^\circ/\text{h} \times 24 \text{ h} \approx 270.72^\circ/\text{day}$

3. ****Period.****
Full turn period (360°):
 $T = \frac{360^\circ}{11.28^\circ/\text{h}} \approx 31.92 \text{ h}$

4. ****Limits.****
- **Equator** ($\lambda = 0^\circ$): $\sin 0^\circ = 0 \rightarrow$ no precession.
- **Pole** ($\lambda = 90^\circ$): $\sin 90^\circ = 1 \rightarrow$ rate $= 15^\circ/\text{h}$, period $= 24 \text{ h}$.
A student using $\cos \lambda$ would be taking the **horizontal** component of Earth's angular velocity, which tilts the local vertical but does **not** produce the observed precession of the swing plane. Hence it is the wrong component for plane precession.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{period_hours} &\approx 31.92 \\ \mathrm{rate_deg_per_day} &\approx 270.72 \\ \mathrm{rate_deg_per_hour} &\approx 11.28 \end{aligned}$$

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:
1. Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
2. Evaluate the precession rate in degrees per hour and degrees per day.
3. Evaluate the full-turn period in hours.
4. What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	270	PASS	-1.0762
rate_deg_per_hour	11.294842	11.25	PASS	-0.044842
period_hours	31.872957	32	PASS	0.127043

Suggested error labels: none

Failed fields: none

Run ID: 43b57069-ea6b-42b1-9339-b978dcf9ad74

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)

2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.

3. **Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.

4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &\approx 271.076197 \\ \mathrm{rate_deg_per_hour} &\approx 11.294842 \\ \mathrm{period_hours} &\approx 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

Step 1: The $\sin \lambda$ factor.
The vertical component of Earth's angular velocity is given by $\Omega_E \sin \lambda$. We need to calculate $\sin(48.85^\circ)$.

Step 2: Calculate $\sin(48.85^\circ)$
 $\sin(48.85^\circ) \approx 0.75$

Step 3: Rate.
 $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. We multiply this by $\sin(48.85^\circ)$ to get the precession rate in degrees per hour.

Step 4: Calculate the precession rate in degrees per hour
 $\text{rate_deg_per_hour} = 15^\circ/\text{h} \times 0.75 = 11.25^\circ/\text{h}$

Step 5: Calculate the precession rate in degrees per day
 $\text{rate_deg_per_day} = 11.25^\circ/\text{h} \times 24 \text{ h} = 270^\circ/\text{day}$

Step 6: Period.
To find the full-turn period, we divide 360° by the precession rate in degrees per hour.

Step 7: Calculate the full-turn period in hours
 $\text{period_hours} = 360^\circ / (11.25^\circ/\text{h}) = 32 \text{ h}$

Step 8: Limits.
At the equator ($\lambda = 0$), $\sin \lambda = 0$, so there is no precession. At the pole ($\lambda = 90^\circ$), the precession rate is $15^\circ/\text{h}$, and the period is 24 h. A student using $\cos \lambda$ takes the horizontal component, which tilts the plane, not precesses it.

The final answer is:
$$\begin{aligned} \mathrm{period_hours} &\approx 32 \\ \mathrm{rate_deg_per_day} &\approx 270 \\ \mathrm{rate_deg_per_hour} &\approx 11.25 \end{aligned}$$

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:

- Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
- Evaluate the precession rate in degrees per hour and degrees per day.
- Evaluate the full-turn period in hours.
- What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	272.04	PASS	0.963803
rate_deg_per_hour	11.294842	11.335	PASS	0.040158
period_hours	31.872957	32.0	PASS	0.127043

Suggested error labels: none

Failed fields: none

Run ID: 981eba36-7bcf-427e-9a83-c05551b36c90

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

- The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
- Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
- Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

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\[\begin{aligned}\mathrm{rate\_deg\_per\_day} &= 271.076197 \\\mathrm{rate\_deg\_per\_hour} &= 11.294842 \\\mathrm{period\_hours} &= 31.872957\end{aligned}\]
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Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

****Derivation****

- **The $\sin \lambda$ factor.**** The Coriolis coupling that rotates the swing plane is the ****vertical**** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- **Rate.**** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) \approx 0.7557$. Rate $= 15 \cdot 0.7557 \approx 11.335^\circ/\text{h}$; $\times 24 = 272.04^\circ/\text{day}$.
- **Period.**** Full turn $= 360^\circ/(11.335^\circ/\text{h}) \approx 32.0 \text{ h}$.
- **Limits.**** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{period\_hours} &= 32.0 \\\mathrm{rate\_deg\_per\_day} &= 272.04 \\\mathrm{rate\_deg\_per\_hour} &= 11.335\end{aligned}\]
```

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:
1. Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
2. Evaluate the precession rate in degrees per hour and degrees per day.
3. Evaluate the full-turn period in hours.
4. What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	267.528	FAIL	-3.5482
rate_deg_per_hour	11.294842	11.147	FAIL	-0.147842
period_hours	31.872957	32.34	FAIL	0.467043

Suggested error labels:
period_hours:MISMATCH;rate_deg_per_day:MISMATCH;rate_deg_per_numeric_mismatch

Failed fields: period_hours;rate_deg_per_day;rate_deg_per_hour

Run ID: 78db6adf-408e-49e7-8f70-5a4c3fe210ad

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
3. **Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &\approx 271.076197 \\ \mathrm{rate_deg_per_hour} &\approx 11.294842 \\ \mathrm{period_hours} &\approx 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

Derivation

1. **The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
2. **Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.7431$. Rate $= 15^\circ/\text{h} \times 0.7431 = 11.147^\circ/\text{h}$; $\times 24 = 267.528^\circ/\text{day}$.
3. **Period.** Full turn $= 360^\circ/11.147^\circ/\text{h} = 32.34 \text{ h}$.
4. **Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{period_hours} &\approx 32.34 \\ \mathrm{rate_deg_per_day} &\approx 267.528 \\ \mathrm{rate_deg_per_hour} &\approx 11.147 \end{aligned}$$

Question

A Foucault pendulum swings at latitude $\lambda = 48.85^\circ$ N (Paris). Treat Earth's rotation rate as 360° per 24 h (ignore the sidereal correction).

Conventions: precession rate of the swing plane $= \Omega_E \sin \lambda$, clockwise viewed from above in the northern hemisphere.

Tasks:

- Derive the $\sin \lambda$ factor from the vertical component of Earth's angular velocity (Coriolis effect on the swing plane).
- Evaluate the precession rate in degrees per hour and degrees per day.
- Evaluate the full-turn period in hours.
- What happens at the equator and at the pole? A student uses $\cos \lambda$ – identify what physical component of $\vec{\Omega}_E$ they took, and why it is the wrong one for plane precession.

Numeric Comparison				
field_name	expected	model_value	match	delta_or_status
rate_deg_per_day	271.076197	271.3	PASS	0.223803
rate_deg_per_hour	11.294842	11.304	PASS	0.009158
period_hours	31.872957	31.84	PASS	-0.032957

Suggested error labels: none

Failed fields: none

Run ID: 87d3ed82-6adb-40f0-8db1-efb66f5207f4

Worked Gold Answer

CM_FOUCAULT_001 — Foucault pendulum: precession at Paris

Given: latitude $\lambda = 48.85^\circ$ N; Earth's rotation 360° per 24 h. Precession rate $= \Omega_E \sin \lambda$, clockwise viewed from above (N hemisphere).

Derivation

- The $\sin \lambda$ factor.** The Coriolis coupling that rotates the swing plane is the **vertical** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- Rate.** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.75299$. Rate $= 15 \cdot 0.75299 = 11.29^\circ/\text{h}$; $\times 24 = 271.08^\circ/\text{day}$.
- Period.** Full turn $= 360^\circ/(11.29^\circ/\text{h}) = 31.87 \text{ h}$.
- Limits.** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component — wrong; it tilts, it does not precess the plane.

FINAL ANSWER

$$\begin{aligned} \mathrm{rate_deg_per_day} &\approx 271.076197 \\ \mathrm{rate_deg_per_hour} &\approx 11.294842 \\ \mathrm{period_hours} &\approx 31.872957 \end{aligned}$$

Reasoning checkpoints: precession $= \Omega_E \sin \lambda$ (vertical component); Paris $\approx 271^\circ/\text{day} \approx 11.3^\circ/\text{h}$, period $\approx 31.9 \text{ h}$; equator none, pole 24 h; $\cos \lambda$ is the horizontal component (tilts, does not precess); clockwise from above in the N hemisphere.

Wrong-answer fingerprints: uses $\cos \lambda \rightarrow$ 'N_SIN_COS_CONFUSED'; misidentifies the rotation component $\rightarrow$ 'C_COMPONENT_MISIDENTIFIED'.

Full Raw Model Response

****Derivation****

- **The $\sin \lambda$ factor.**** The Coriolis coupling that rotates the swing plane is the ****vertical**** component of Earth's angular velocity, $\Omega_E \sin \lambda$. (The horizontal component $\Omega_E \cos \lambda$ tilts the local vertical; it does not precess the plane.)
- **Rate.**** $\Omega_E = 360^\circ/24 \text{ h} = 15^\circ/\text{h}$. $\sin(48.85^\circ) = 0.7536$. Rate $= 15 \cdot 0.7536 = 11.304^\circ/\text{h}$; $\times 24 = 271.3^\circ/\text{day}$.
- **Period.**** Full turn $= 360^\circ/(11.304^\circ/\text{h}) = 31.84 \text{ h}$.
- **Limits.**** Equator ($\lambda = 0$): $\sin \lambda = 0 \rightarrow$ no precession. Pole ($\lambda = 90^\circ$): $15^\circ/\text{h} \rightarrow 24 \text{ h}$ period. A student using $\cos \lambda$ takes the horizontal component – wrong; it tilts, it does not precess the plane.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{period_hours} &\approx 31.84 \\ \mathrm{rate_deg_per_day} &\approx 271.3 \\ \mathrm{rate_deg_per_hour} &\approx 11.304 \end{aligned}$$

Question

The CMB today has $T_0 = 2.725\text{ K}$. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:
1. How does the photon temperature scale with redshift? Evaluate T at recombination in K.
2. How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
3. How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
4. Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	2999.025	PASS	-1.2
log10_number_ratio	9.125362	9.1242	PASS	-0.001162
log10_energy_ratio	12.167149	12.1656	PASS	-0.001549

Suggested error labels: none

Failed fields: none

Run ID: f7f12cfe-6b34-4c5b-8af3-5337de7fb235

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725\text{ K}$, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

1. **Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.

2. **Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.

3. **Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.

4. **Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 3000.225 \\ \mathrm{\log_{10}\,number_ratio} &= 9.125362 \\ \mathrm{\log_{10}\,energy_ratio} &= 12.167149 \end{aligned}$$

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000\text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ ``N_ENERGY_NUMBER_SCALING_CONFLATED``; invokes rethermalization $\rightarrow$ ``C_RETHERMALIZATION_INVOKED``.

Full Raw Model Response

****Derivation****

1. ****Temperature.**** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = 2999.025\text{ K}$.

2. ****Number density.**** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = 3 * 3.0414 = 9.1242$.

3. ****Energy density.**** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = 4 * 3.0414 = 12.1656$. The extra factor of $(1+z)$ over the number density is the ****per-photon energy redshift**** — each photon's energy scales as $(1+z)$.

4. ****Blackbody preserved.**** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 2999.025 \\ \mathrm{\log_{10}\,energy_ratio} &= 12.1656 \\ \mathrm{\log_{10}\,number_ratio} &= 9.1242 \end{aligned}$$

Question

The CMB today has $T_0 = 2.725$ K. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

- Tasks:
- How does the photon temperature scale with redshift? Evaluate T at recombination in K.
 - How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
 - How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
 - Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	3000.225	PASS	0
log10_number_ratio	9.125362	9.12417	PASS	-0.001192
log10_energy_ratio	12.167149	12.16556	PASS	-0.001589

Suggested error labels: none

Failed fields: none

Run ID: 45f12cf7-3772-421a-8998-b0b7ff23252b

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725$ K, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

- Derivation**
- Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.
 - Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
 - Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
 - Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

```
\[
\begin{aligned}
\mathrm{T_{rec}\,K} &= 3000.225 \\
\mathrm{\log_{10}\,number\_ratio} &= 9.125362 \\
\mathrm{\log_{10}\,energy\_ratio} &= 12.167149
\end{aligned}
\]
```

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000\text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ ``N_ENERGY_NUMBER_SCALING_CONFLATED``; invokes rethermalization $\rightarrow$ ``C_RETHERMALIZATION_INVOKED``.

Full Raw Model Response

```
**Derivation**
1. **Temperature.**  $T \propto (1+z)$ :  $T_{\text{rec}} = 2.725 \cdot 1101 = 3000.225\text{ K}$ .
2. **Number density.**  $n \propto (1+z)^3$ :  $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = 3 \cdot 3.04139 = 9.12417$ .
3. **Energy density.**  $u \propto (1+z)^4$ :  $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = 4 \cdot 3.04139 = 12.16556$ . The extra factor of  $(1+z)$  over the number density is the per-photon energy redshift — each photon's energy scales as  $(1+z)$ .
4. **Blackbody preserved.**  $T \propto 1/a$  keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at  $T$  yields a Planck spectrum at  $T/(1+z)$ . No scattering or rethermalization is required.

**FINAL ANSWER**

\[
\begin{aligned}
\mathrm{T_{rec}\,K} &= 3000.225 \\
\mathrm{\log_{10}\,energy\_ratio} &= 12.16556 \\
\mathrm{\log_{10}\,number\_ratio} &= 9.12417
\end{aligned}
\]
```

Question

The CMB today has $T_0 = 2.725\text{ K}$. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:
1. How does the photon temperature scale with redshift? Evaluate T at recombination in K.
2. How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
3. How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
4. Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	2999.925	PASS	-0.3
log10_number_ratio	9.125362	9.125	PASS	-0.000362
log10_energy_ratio	12.167149	12.167	PASS	-0.000149

Suggested error labels: noneFailed fields: noneRun ID: c15d63d9-68db-48a3-85f1-9600fff091ee

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725\text{ K}$, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

1. **Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.
2. **Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
3. **Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
4. **Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 3000.225 \\ \mathrm{\log_{10}\,number_ratio} &= 9.125362 \\ \mathrm{\log_{10}\,energy_ratio} &= 12.167149 \end{aligned}$$

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000\text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ 'N_ENERGY_NUMBER_SCALING_CONFLATED'; invokes rethermalization $\rightarrow$ 'C_RETHERMALIZATION_INVOKED'.

Full Raw Model Response

****Derivation****

1. ****Temperature.**** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot (1 + 1100) = 2.725 \cdot 1101 = \mathbf{2999.925\text{ K}}$.

2. ****Number density.**** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) \approx 3 \cdot 3.0418 = \mathbf{9.125}$.

3. ****Energy density.**** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) \approx 4 \cdot 3.0418 = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.

4. ****Blackbody preserved.**** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 2999.925 \\ \mathrm{\log_{10}\,energy_ratio} &= 12.167 \\ \mathrm{\log_{10}\,number_ratio} &= 9.125 \end{aligned}$$

Question

The CMB today has $T_0 = 2.725$ K. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:

- How does the photon temperature scale with redshift? Evaluate T at recombination in K.
- How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
- How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
- Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	3000.0	PASS	-0.225
log10_number_ratio	9.125362	9.124	PASS	-0.001362
log10_energy_ratio	12.167149	12.166	PASS	-0.001149

Suggested error labels: none

Failed fields: none

Run ID: 33b7ba26-8ce1-4c4c-aadf-b13a0800ceed

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725$ K, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

- Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.
- Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
- Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
- Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

```
\begin{aligned}\mathrm{T_{rec}\_K} &= 3000.225 \\\mathrm{log10\_number\_ratio} &= 9.125362 \\\mathrm{log10\_energy\_ratio} &= 12.167149\end{aligned}
```

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000$ K; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ ``N_ENERGY_NUMBER_SCALING_CONFLATED``; invokes rethermalization $\rightarrow$ ``C_RETHERMALIZATION_INVOKED``.

Full Raw Model Response

****Derivation****

- **Temperature.****
The photon temperature scales inversely with the cosmic scale factor:
 $(T \propto 1/a \propto (1+z)^{-1})$.
 $T_{\text{rec}} = T_0 (1+z_{\text{rec}}) = 2.725 \times 1101 \approx 3000.0$ K.
- **Number density.****
The number density of photons scales as $(n \propto a^{-3} \propto (1+z)^3)$.
 $\log_{10}(n_{\text{rec}}/n_0) = 3 \log_{10}(1101) \approx 3 \times 3.0414 = 9.124$.
- **Energy density.****
The energy density scales as $(u \propto a^{-4} \propto (1+z)^4)$.
 $\log_{10}(u_{\text{rec}}/u_0) = 4 \log_{10}(1101) \approx 4 \times 3.0414 = 12.166$.
The extra factor of $(1+z)$ relative to the number density arises because each photon's energy is redshifted as $(E \propto 1/(1+z))$. Thus, while the number of photons falls as $((1+z)^{-3})$, the energy per photon falls as $((1+z)^{-1})$, giving an overall energy density scaling of $((1+z)^{-4})$.
- **Blackbody preserved.****
The cosmic microwave background remains a perfect blackbody because the expansion of the universe is adiabatic for a photon gas. In an adiabatic expansion, the shape of the Planck spectrum is invariant: a Planck distribution at temperature (T) in an expanding universe redshifts to a Planck distribution at temperature $(T/(1+z))$. No scattering or rethermalization events are required to maintain the blackbody form.

****FINAL ANSWER****

```
\begin{aligned}\mathrm{T_{rec}\_K} &= 3000.0 \\\mathrm{log10\_energy\_ratio} &= 12.166 \\\mathrm{log10\_number\_ratio} &= 9.124\end{aligned}
```

Question

The CMB today has $T_0 = 2.725$ K. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:
1. How does the photon temperature scale with redshift? Evaluate T at recombination in K.
2. How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
3. How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
4. Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	29885.275	FAIL	26885.1
log10_number_ratio	9.125362	6.041	FAIL	-3.08436
log10_energy_ratio	12.167149	8.083	FAIL	-4.08415

Suggested error labels:
T_rec_K:MISMATCH;log10_energy_ratio:MISMATCH;log10_number_ratio:MISMATCH;numeric_mismatch

Failed fields: T_rec_K;log10_energy_ratio;log10_number_ratio

Run ID: a3752b06-d12a-4b6f-b533-06abdc5a4a2d

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725$ K, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

1. **Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.

2. **Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.

3. **Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.

4. **Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 3000.225 \\ \mathrm{\log_{10}\,number_ratio} &= 9.125362 \\ \mathrm{\log_{10}\,energy_ratio} &= 12.167149 \end{aligned}$$

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000$ K; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ `N\_ENERGY\_NUMBER\_SCALING\_CONFLATED`; invokes rethermalization $\rightarrow$ `C\_RETHERMALIZATION\_INVOKED`.

Full Raw Model Response

****Derivation****

1. ****Temperature.**** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = 29885.275$ K.

2. ****Number density.**** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = 6.041$.

3. ****Energy density.**** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = 8.083$. The extra factor of $(1+z)$ over the number density is the per-photon energy redshift — each photon's energy scales as $(1+z)$.

4. ****Blackbody preserved.**** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{T_{rec}\,K} &= 29885.275 \\ \mathrm{\log_{10}\,energy_ratio} &= 8.083 \\ \mathrm{\log_{10}\,number_ratio} &= 6.041 \end{aligned}$$

Question

The CMB today has $T_0 = 2.725\text{ K}$. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:

- How does the photon temperature scale with redshift? Evaluate T at recombination in K.
- How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
- How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
- Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	3000.225	PASS	0
log10_number_ratio	9.125362	9.1257	PASS	0.000338
log10_energy_ratio	12.167149	12.1676	PASS	0.000451

Suggested error labels: none

Failed fields: none

Run ID: ea1562a8-ce25-46ef-96c3-4ea66084be94

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725\text{ K}$, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

- Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.
- Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
- Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
- Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{T_{rec}\,K} &= 3000.225 \\\mathrm{\log_{10}\,number\_ratio} &= 9.125362 \\\mathrm{\log_{10}\,energy\_ratio} &= 12.167149\end{aligned}\]
```

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000\text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ ``N_ENERGY_NUMBER_SCALING_CONFLATED``; invokes rethermalization $\rightarrow$ ``C_RETHERMALIZATION_INVOKED``.

Full Raw Model Response

****Derivation****

1. **Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \times (1 + 1100) = 2.725 \times 1101 = 3000.225\text{ K}$.

2. **Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) \approx 3 \times 3.0419 = 9.1257$.

3. **Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) \approx 4 \times 3.0419 = 12.1676$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.

4. **Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{T_{rec}\,K} &= 3000.225 \\\mathrm{\log_{10}\,energy\_ratio} &= 12.1676 \\\mathrm{\log_{10}\,number\_ratio} &= 9.1257\end{aligned}\]
```

Question

The CMB today has $T_0 = 2.725\text{ K}$. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:

- How does the photon temperature scale with redshift? Evaluate T at recombination in K.
- How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
- How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
- Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	3000.25	PASS	0.025
log10_number_ratio	9.125362	3.030	FAIL	-6.09536
log10_energy_ratio	12.167149	4.040	FAIL	-8.12715

Suggested error labels:
log10_energy_ratio:MISMATCH;log10_number_ratio:MISMATCH;numeric_mismatch

Failed fields: log10_energy_ratio;log10_number_ratio

Run ID: 5bde406e-bcbd-4d75-9974-030635a1f45a

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725\text{ K}$, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

- Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2\text{ K}}$.
- Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
- Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
- Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{T_{rec}\,K} &= 3000.225 \\\mathrm{\log_{10}\,number\_ratio} &= 9.125362 \\\mathrm{\log_{10}\,energy\_ratio} &= 12.167149\end{aligned}\]
```

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000\text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ `N\_ENERGY\_NUMBER\_SCALING\_CONFLATED`; invokes rethermalization $\rightarrow$ `C\_RETHERMALIZATION\_INVOKED`.

Full Raw Model Response

****Derivation****

- **Temperature.**** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.25\text{ K}}$.
- **Number density.**** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.
- **Energy density.**** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.
- **Blackbody preserved.**** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{T_{rec}\,K} &= 3000.25 \\\mathrm{\log_{10}\,energy\_ratio} &= 12.167 \\\mathrm{\log_{10}\,number\_ratio} &= 9.125\end{aligned}\]
```


Question

The CMB today has $T_0 = 2.725$ K. Recombination occurred at redshift $z = 1100$.

Conventions: adiabatic expansion of a blackbody photon gas; report $\log_{10}$ for the density ratios (recombination relative to today).

Tasks:
1. How does the photon temperature scale with redshift? Evaluate T at recombination in K.
2. How does the photon NUMBER density scale? Give $\log_{10}$ of the ratio $n(z_{\text{rec}})/n_0$.
3. How does the photon ENERGY density scale? Give $\log_{10}$ of the ratio, and explain where the extra factor of $(1+z)$ relative to number density comes from.
4. Why does the CMB remain a perfect blackbody as it redshifts, without any rethermalization?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
T_rec_K	3000.225	2999.725	PASS	-0.5
log10_number_ratio	9.125362	3.0414	FAIL	-6.08396
log10_energy_ratio	12.167149	4.0552	FAIL	-8.11195

Suggested error labels:
log10_energy_ratio:MISMATCH;log10_number_ratio:MISMATCH;numeric_mismatch

Failed fields: log10_energy_ratio;log10_number_ratio

Run ID: 2d1f22bc-291c-4722-88c7-cc1fed3ce53d

Worked Gold Answer

COS_CMB_001 — CMB scaling: temperature, number, and energy densities

Given: $T_0 = 2.725$ K, recombination at $z = 1100$. Report $\log_{10}$ of density ratios (recombination / today).

Derivation

1. **Temperature.** $T \propto (1+z)$: $T_{\text{rec}} = 2.725 \cdot 1101 = \mathbf{3000.2 \text{ K}}$.

2. **Number density.** $n \propto (1+z)^3$: $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = \mathbf{9.125}$.

3. **Energy density.** $u \propto (1+z)^4$: $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = \mathbf{12.167}$. The extra factor of $(1+z)$ over the number density is the **per-photon energy redshift** — each photon's energy scales as $(1+z)$.

4. **Blackbody preserved.** $T \propto 1/a$ keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at T yields a Planck spectrum at $T/(1+z)$. No scattering or rethermalization is required.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{T_{rec}\,K} &= 3000.225 \\\mathrm{\log_{10}\,number\_ratio} &= 9.125362 \\\mathrm{\log_{10}\,energy\_ratio} &= 12.167149\end{aligned}\]
```

Reasoning checkpoints: $T \propto (1+z) \rightarrow 3000 \text{ K}$; $n \propto (1+z)^3 \rightarrow \log 9.13$; $u \propto (1+z)^4 \rightarrow \log 12.17$, extra $(1+z)$ from per-photon redshift; blackbody preserved (no rethermalization); does not use $(1+z)^3$ for energy density.

Wrong-answer fingerprints: energy scaled as $(1+z)^3$ / conflated with number $\rightarrow$ 'N_ENERGY_NUMBER_SCALING_CONFLATED'; invokes rethermalization $\rightarrow$ 'C_RETHERMALIZATION_INVOKED'.

Full Raw Model Response

```
**Derivation**
1. **Temperature.**  $T \propto (1+z)$ :  $T_{\text{rec}} = 2.725 \cdot 1101 = 2999.725 \text{ K}$ .
2. **Number density.**  $n \propto (1+z)^3$ :  $\log_{10}(n_{\text{rec}}/n_0) = 3 \cdot \log_{10}(1101) = 3.0414$ .
3. **Energy density.**  $u \propto (1+z)^4$ :  $\log_{10}(u_{\text{rec}}/u_0) = 4 \cdot \log_{10}(1101) = 4.0552$ . The extra factor of  $(1+z)$  over the number density is the per-photon energy redshift — each photon's energy scales as  $(1+z)$ .
4. **Blackbody preserved.**  $T \propto 1/a$  keeps the Planck spectrum form-invariant: redshifting a Planck spectrum at  $T$  yields a Planck spectrum at  $T/(1+z)$ . No scattering or rethermalization is required.
**FINAL ANSWER**

\[\begin{aligned}\mathrm{T_{rec}\,K} &= 2999.725 \\\mathrm{\log_{10}\,energy\_ratio} &= 4.0552 \\\mathrm{\log_{10}\,number\_ratio} &= 3.0414\end{aligned}\]
```

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	7.356	FAIL	-81.2642
f_orbital_hz	219.802347	9.13e6	FAIL	9.12978e+06
f_gw_hz	439.604695	1.826e7	FAIL	1.82596e+07

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;r_isco_km:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz;r_isco_km

Run ID: 0891d888-672d-4f87-bc31-5a40a8dfdd32

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4$ m $\Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{r_{isco}\_km} &\quad \&= 88.620232 \\\mathrm{f_{orbital}\_hz} &\quad \&= 219.802347 \\\mathrm{f_{gw}\_hz} &\quad \&= 439.604695\end{aligned}\]
```

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, ****ISCO $6GM/c^2$ **** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 6.674e-11 \cdot 1.989e30 / (2.998e8)^2 \approx 2.00e10$ m³/s² ≈ 1.226 km: horizon 2.452 km, photon sphere 3.678 km, ****r_ISCO = 6·1.226 ≈ 7.356 km****.
- **Orbital frequency.**** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 6.674e-11 \cdot 1.989e30 = 1.327e41$ m³/s², $r_{\text{ISCO}} = 7356$ m $\Rightarrow$ $f_{\text{orb}} = (1/(2\pi)) \cdot \sqrt{(1.327e41 / 7356^3)} \approx (1/(2\pi)) \cdot \sqrt{(1.327e41 / 400e9)} \approx (1/(2\pi)) \cdot \sqrt{(3.318e14)} \approx (1/(2\pi)) \cdot 5.76e7 \approx 9.13e6$ Hz.
- **GW frequency.**** Dominant emission at twice the orbital frequency: ****f_GW = 2·9.13e6 = 1.826e7 Hz****.
- **Interpretation.**** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at ****lower**** frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{f_{gw}\_hz} &\quad \&= 1.826e7 \\\mathrm{f_{orbital}\_hz} &\quad \&= 9.13e6 \\\mathrm{r_{isco}\_km} &\quad \&= 7.356\end{aligned}\]
```

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	88.596	PASS	-0.024232
f_orbital_hz	219.802347	2014.5	FAIL	1794.7
f_gw_hz	439.604695	4029.0	FAIL	3589.4

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH; numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz

Run ID: a043fef4-92a9-4e39-8b94-4099ce5d7bd8

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4$ m $\Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{r_{isco}\_km} &= 88.620232 \\\mathrm{f_{orbital}\_hz} &= 219.802347 \\\mathrm{f_{gw}\_hz} &= 439.604695\end{aligned}\]
```

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

thought

****Derivation****

- **Characteristic radii**** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, ****ISCO $6GM/c^2$ **** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.766$ km: horizon 29.532 km, photon sphere 44.298 km, ****r_ISCO = 6·14.766 ≈ 88.596 km****.
- **Orbital frequency.**** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.32712 \times 10^{21}$ m³/s², $r_{\text{ISCO}} = 88596$ m $\Rightarrow$ ****f_orb = 2014.5 Hz****.
- **GW frequency.**** Dominant emission at twice the orbital frequency: ****f_GW = 2·2014.5 = 4029.0 Hz****.
- **Interpretation.**** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at ****lower**** frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{f_{gw}\_hz} &= 4029.0 \\\mathrm{f_{orbital}\_hz} &= 2014.5 \\\mathrm{r_{isco}\_km} &= 88.596\end{aligned}\]
```

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	88.6	PASS	-0.020232
f_orbital_hz	219.802347	197.3	FAIL	-22.5023
f_gw_hz	439.604695	394.6	FAIL	-45.0047

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH; numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz

Run ID: e4d58dfa-4086-43b4-a5bf-1f9bec6c2599

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

$$\begin{aligned} \mathrm{r_isco_km} &\approx 88.620232 \\ \mathrm{f_orbital_hz} &\approx 219.802347 \\ \mathrm{f_gw_hz} &\approx 439.604695 \end{aligned}$$

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, ****ISCO $6GM/c^2$ **** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.766$ km: horizon 29.532 km, photon sphere 44.298 km, ****r_ISCO = 6·14.766 ≈ 88.596 km****.
- **Orbital frequency.**** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.32712 \times 10^{21} \text{ m}^3/\text{s}^2$, $r_{\text{ISCO}} = 88596 \text{ m} \Rightarrow$ ****f_orb = 197.3 Hz****.
- **GW frequency.**** Dominant emission at twice the orbital frequency: ****f_GW = 2·197.3 = 394.6 Hz****.
- **Interpretation.**** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at ****lower**** frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{f_gw_hz} &\approx 394.6 \\ \mathrm{f_orbital_hz} &\approx 197.3 \\ \mathrm{r_isco_km} &\approx 88.6 \end{aligned}$$

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	88.5	PASS	-0.120232
f_orbital_hz	219.802347	1.47×10^{-8}	FAIL	
f_gw_hz	439.604695	2.94×10^{-8}	FAIL	

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz

Run ID: 40dd5ad7-5daa-4765-ad81-b279dda17267

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **$r_{\text{ISCO}} = 6 \cdot 14.77 \approx 88.62$ km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow \mathbf{f_{orb} = 219.80 \text{ Hz}}$.
- GW frequency.** Dominant emission at twice the orbital frequency: **$f_{\text{GW}} = 2 \cdot 219.80 = 439.60 \text{ Hz}$** .
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

$$\begin{aligned} \mathrm{r_{isco_km}} &\approx 88.620232 \\ \mathrm{f_{orbital_hz}} &\approx 219.802347 \\ \mathrm{f_{gw_hz}} &\approx 439.604695 \end{aligned}$$

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of (GM/c^2)):
 - Event horizon: $(r_{\text{H}} = 2 \sqrt{GM/c^2}) \rightarrow **2** (GM/c^2)$.
 - Photon sphere: $(r_{\text{ph}} = 3 \sqrt{GM/c^2}) \rightarrow **3** (GM/c^2)$.
 - ISCO: $(r_{\text{ISCO}} = 6 \sqrt{GM/c^2}) \rightarrow **6** (GM/c^2)$.

Compute (GM/c^2) for $(M = 10 M_\odot)$:

$$GM = G \times 10 M_\odot = 6.674 \times 10^{-11} \times 1.989 \times 10^{30} \approx 1.327 \times 10^{21} \text{ kg m}^3 \text{ s}^{-2}$$
$$\frac{GM}{c^2} = \frac{1.327 \times 10^{21}}{(2.998 \times 10^8)^2} \approx \frac{1.327 \times 10^{21}}{8.988 \times 10^{16}} \approx 1.475 \times 10^4 \text{ m} \approx 14.75 \text{ km}$$

Hence:

 - Horizon: $(2 \times 14.75 \approx 29.5 \text{ km})$.
 - Photon sphere: $(3 \times 14.75 \approx 44.3 \text{ km})$.
 - ISCO: $(6 \times 14.75 \approx 88.5 \text{ km})$.

So **$r_{\text{ISCO}} \approx 88.5 \text{ km}$** .
- **Orbital frequency**** at the ISCO:

$$f_{\text{orb}} = \frac{\Omega}{2\pi}, \quad \Omega = \sqrt{\frac{GM}{r_{\text{ISCO}}^3}}$$

With $(r_{\text{ISCO}} = 6 \sqrt{GM/c^2})$:

$$\begin{aligned} \Omega &= \sqrt{\frac{GM}{(6 \sqrt{GM/c^2})^3}} \\ &= \sqrt{\frac{GM}{216 \sqrt{GM/c^2}}} \\ &= \frac{\sqrt{GM/c^3}}{\sqrt{216} M} \end{aligned}$$

Therefore,

$$f_{\text{orb}} = \frac{\Omega}{2\pi} = \frac{\sqrt{GM/c^3}}{2\pi \sqrt{216} M}$$

Insert $(c = 2.998 \times 10^8 \text{ m/s})$ and $(M = 10 M_\odot = 1.989 \times 10^{31} \text{ kg})$:

$$f_{\text{orb}} = \frac{(2.998 \times 10^8)^3}{2\pi \sqrt{216} (1.989 \times 10^{31})} \approx \frac{2.697 \times 10^{24}}{2\pi \times 14.696 \times 1.989 \times 10^{31}} \approx \frac{2.697 \times 10^{24}}{1.835 \times 10^{32}} \approx 1.47 \times 10^{-8} \text{ Hz}$$

$f_{\text{orbital_hz}} \approx 1.47 \times 10^{-8} \text{ Hz}$.
- **Gravitational-wave frequency**** (dominant emission at twice the orbital frequency):

$$f_{\text{GW}} = 2 f_{\text{orb}} \approx 2 \times 1.47 \times 10^{-8} \approx 2.94 \times 10^{-8} \text{ Hz}$$

$f_{\text{gw_hz}} \approx 2.94 \times 10^{-8} \text{ Hz}$.
- **Interpretation**:**

No stable circular orbit exists inside the ISCO; once the binary reaches this radius the orbit becomes plunging, ending the quasi-periodic inspiral. Consequently the characteristic “chirp” of gravitational waves ceases near the ISCO frequency. Because the ISCO radius scales as $(r_{\text{ISCO}} \propto M)$ and the orbital frequency scales as $(f_{\text{ISCO}} \propto M^{-1})$, heavier black-hole binaries merge at lower GW frequencies (i.e., $(f_{\text{ISCO}} \propto 1/M)$).

****FINAL ANSWER****

$$\begin{aligned} \end{aligned}$$

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	1.772×10^6	FAIL	
f_orbital_hz	219.802347	1.24×10^4	FAIL	
f_gw_hz	439.604695	2.48×10^4	FAIL	

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;r_isco_km:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz;r_isco_km

Run ID: 12ba6901-ab09-4608-812d-243a3ba8cb47

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}})^3}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

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\[\begin{aligned}\mathrm{r\_isco\_km} \quad & \approx 88.620232 \quad \backslash \backslash \\ \mathrm{f\_orbital\_hz} \quad & \approx 219.802347 \quad \backslash \backslash \\ \mathrm{f\_gw\_hz} \quad & \approx 439.604695 \quad \backslash \backslash \\ \end{aligned}\]
```

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

Step 1: Calculate GM/c^2 in km
 $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx (6.674 \times 10^{-11}) \cdot (1.989 \times 10^{30}) / (2.998 \times 10^8)^2 \approx 2.953 \times 10^9 \text{ m}^2/\text{s}^2$
 GM/c^2 in km = $2.953 \times 10^9 \text{ m}^2/\text{s}^2 \approx 2.953 \times 10^9 \text{ m}^2/\text{s}^2 / (1000 \text{ m}/\text{km})^2 \approx 2.953 \times 10^5 \text{ km}^2/\text{s}^2$
 GM/c^2 in km = $\sqrt{(2.953 \times 10^5 \text{ km}^2/\text{s}^2)} \approx 542.5 \text{ km}$

Step 2: Calculate the radii of the event horizon, the photon sphere, and the ISCO
Event horizon: $2GM/c^2 = 2 \cdot 2.953 \times 10^5 \text{ km} \approx 5.906 \times 10^5 \text{ km}$
Photon sphere: $3GM/c^2 = 3 \cdot 2.953 \times 10^5 \text{ km} \approx 8.859 \times 10^5 \text{ km}$
ISCO: $6GM/c^2 = 6 \cdot 2.953 \times 10^5 \text{ km} \approx 1.772 \times 10^6 \text{ km}$

Step 3: Calculate the orbital frequency at the ISCO
 $GM = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2 \cdot 1.989 \times 10^{30} \text{ kg} \approx 1.327 \times 10^{20} \text{ m}^3/\text{s}^2$
 $r_{\text{ISCO}} = 1.772 \times 10^6 \text{ km} = 1.772 \times 10^9 \text{ m}$
 $f_{\text{orb}} = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}})^3} = (1/2\pi)\sqrt{(1.327 \times 10^{20} \text{ m}^3/\text{s}^2 / (1.772 \times 10^9 \text{ m})^3)} \approx 1.24 \times 10^{-4} \text{ Hz}$

Step 4: Calculate the GW frequency at the ISCO
 $f_{\text{GW}} = 2 \cdot f_{\text{orb}} \approx 2 \cdot 1.24 \times 10^{-4} \text{ Hz} \approx 2.48 \times 10^{-4} \text{ Hz}$

Step 5: Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales with M
The GW signal 'shuts off' near this frequency because no stable circular orbit exists inside the ISCO, causing the inspiral to transition to a plunge/merger. The ISCO frequency scales with M as $f_{\text{ISCO}} \propto 1/M$.

The final answer is:

```
\[\begin{aligned}\mathrm{f\_gw\_hz} \quad & \approx 2.48 \times 10^{-4} \quad \backslash \backslash \\ \mathrm{f\_orbital\_hz} \quad & \approx 1.24 \times 10^{-4} \quad \backslash \backslash \\ \mathrm{r\_isco\_km} \quad & \approx 1.772 \times 10^6 \quad \backslash \backslash \\ \end{aligned}\]
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Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	$8.86 \times 10^3 \text{ km}$	FAIL	
f_orbital_hz	219.802347	$1.76 \times 10^{-4} \text{ Hz}$	FAIL	
f_gw_hz	439.604695	$3.52 \times 10^{-4} \text{ Hz}$	FAIL	

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;r_isco_km:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz;r_isco_km

Run ID: 6a0624c3-57f8-4327-8206-731c8c16e32e

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

$$\begin{aligned} \mathrm{r_isco_km} &\approx 88.620232 \\ \mathrm{f_orbital_hz} &\approx 219.802347 \\ \mathrm{f_gw_hz} &\approx 439.604695 \end{aligned}$$

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of GM/c^2):
 - event horizon: $\backslash(2GM/c^2\backslash)$
 - photon sphere: $\backslash(3GM/c^2\backslash)$
 - **ISCO****: $\backslash(6GM/c^2\backslash)$ (for Schwarzschild, non-spinning case)
 - $\backslash(GM/c^2 = G \cdot 10 M_\odot / c^2 = 6.674 \times 10^{-11} \cdot 10 \cdot 1.989 \times 10^{30} / (2.998 \times 10^8)^2\backslash)$
 $\backslash(= 1.476 \times 10^3 \text{ km}\backslash)$
 - horizon: $\backslash(2 \cdot 1.476 \times 10^3 = 2.952 \times 10^3 \text{ km}\backslash)$
 - photon sphere: $\backslash(3 \cdot 1.476 \times 10^3 = 4.428 \times 10^3 \text{ km}\backslash)$
 - **r_ISCO = 6·1.476 × 10³ ≈ 8.856 × 10³ km****
- **Orbital frequency.****
 - $\backslash(GM = 6.674 \times 10^{-11} \cdot 1.989 \times 10^{31} = 1.329 \times 10^{20} \text{ m}^3/\text{s}^2\backslash)$
 - $\backslash(r_{\text{ISCO}} = 8.856 \times 10^6 \text{ m}\backslash)$
 - $\backslash(\Omega = \sqrt{GM/r_{\text{ISCO}}^3} = \sqrt{1.329 \times 10^{20} / (8.856 \times 10^6)^3} = 1.105 \times 10^{-3} \text{ rad/s}\backslash)$
 - **f_orb = $\Omega/2\pi = 1.105 \times 10^{-3} / 6.283 \approx 1.76 \times 10^{-4}$ Hz****
- **GW frequency.****
 - **f_GW = 2·1.76 × 10⁻⁴ = 3.52 × 10⁻⁴ Hz****
- **Interpretation.****
 - No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there – the quasi-periodic chirp "shuts off."
 - Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at ****lower**** frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{f_gw_hz} &\approx 3.52 \times 10^{-4} \text{ Hz} \\ \mathrm{f_orbital_hz} &\approx 1.76 \times 10^{-4} \text{ Hz} \\ \mathrm{r_isco_km} &\approx 8.86 \times 10^3 \text{ km} \end{aligned}$$

Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $\dot{M} = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	17.7	FAIL	-70.9202
f_orbital_hz	219.802347	129	FAIL	-90.8023
f_gw_hz	439.604695	258	FAIL	-181.605

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;r_isco_km:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz;r_isco_km

Run ID: afe1c751-3c98-4578-ae5-2bc6938397b8

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{r_{isco}\_km} & \approx 88.620232 \\\mathrm{f_{orbital}\_hz} & \approx 219.802347 \\\mathrm{f_{gw}\_hz} & \approx 439.604695\end{aligned}\]
```

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, ****ISCO $6GM/c^2$ **** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 2.95$ km: horizon 5.9 km, photon sphere 8.85 km, ****r_ISCO = 6·2.95 ≈ 17.7 km****.
- **Orbital frequency.**** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 6.674 \times 10^{-11} \times 10 \times 1.989 \times 10^{30} \text{ m}^3/\text{s}^2$, $r_{\text{ISCO}} = 17700 \text{ m} \Rightarrow$ ****f_orb = 129 Hz****.
- **GW frequency.**** Dominant emission at twice the orbital frequency: ****f_GW = 2·129 = 258 Hz****.
- **Interpretation.**** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at ****lower**** frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{r_{gw}\_hz} & \approx 258 \\\mathrm{f_{orbital}\_hz} & \approx 129 \\\mathrm{r_{isco}\_km} & \approx 17.7\end{aligned}\]
```


Question

A test particle orbits a Schwarzschild black hole of mass $M = 10 M_\odot$. Use $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$ SI, $c = 2.998 \times 10^8$ m/s.

Conventions: Schwarzschild coordinates; orbital frequency means $\Omega = d\phi/dt$ with respect to coordinate time (frequency measured at infinity). For circular equatorial orbits in Schwarzschild, $\Omega^2 = GM/r^3$ holds EXACTLY in these coordinates.

Tasks:

- Give the radii of the event horizon, the photon sphere, and the ISCO in units of GM/c^2 , and evaluate the ISCO radius in km.
- Evaluate the orbital frequency $f = \Omega/2\pi$ at the ISCO in Hz.
- The dominant gravitational-wave emission is at twice the orbital frequency. Evaluate f_{GW} at the ISCO.
- Explain why a compact binary's GW signal 'shuts off' near this frequency and how the ISCO frequency scales

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
r_isco_km	88.620232	8859.6	FAIL	8770.98
f_orbital_hz	219.802347	1.554×10^{-3}	FAIL	
f_gw_hz	439.604695	3.108×10^{-3}	FAIL	

Suggested error labels:
f_gw_hz:MISMATCH;f_orbital_hz:MISMATCH;r_isco_km:MISMATCH;numeric_mismatch

Failed fields: f_gw_hz;f_orbital_hz;r_isco_km

Run ID: ab92bfee-92d9-4fbc-b33c-4d25777badf5

Worked Gold Answer

GR_ISCO_001 — Schwarzschild ISCO: radius and GW frequency

Given: $M = 10 M_\odot$, $M_\odot = 1.989 \times 10^{30}$ kg, $G = 6.674 \times 10^{-11}$, $c = 2.998 \times 10^8$ m/s. Exact Schwarzschild identity $\Omega^2 = GM/r^3$.

Derivation

- Characteristic radii** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, **ISCO $6GM/c^2$** — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 14.77$ km: horizon 29.54 km, photon sphere 44.31 km, **r_ISCO = 6·14.77 ≈ 88.62 km**.
- Orbital frequency.** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.3276 \times 10^{21}$, $r_{\text{ISCO}} = 8.862 \times 10^4 \text{ m} \Rightarrow$ **f_orb = 219.80 Hz**.
- GW frequency.** Dominant emission at twice the orbital frequency: **f_GW = 2·219.80 = 439.60 Hz**.
- Interpretation.** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at **lower** frequency ($f_{\text{ISCO}} \propto 1/M$).

FINAL ANSWER

```
\[
\begin{aligned}
\mathrm{r_{isco}\_km} & \quad \&= 88.620232 \\
\mathrm{f_{orbital}\_hz} & \quad \&= 219.802347 \\
\mathrm{f_{gw}\_hz} & \quad \&= 439.604695
\end{aligned}
\]
```

Reasoning checkpoints: horizon 2 / photon sphere 3 / ISCO 6 GM/c^2 all distinguished; $r_{\text{ISCO}} \approx 88.6$ km; uses the exact $\Omega^2 = GM/r^3$ (no ad-hoc correction); $f_{\text{orb}} \approx 220$, $f_{\text{GW}} \approx 440$; states $f_{\text{ISCO}} \propto 1/M$.

Wrong-answer fingerprints: ISCO given as 2 or 3 $GM/c^2 \rightarrow$ 'N_RADIUS_CONFUSION'; $f_{\text{GW}} = f_{\text{orb}}$ (factor-2 dropped) $\rightarrow$ 'N_GW_FACTOR2_DROPPED'; adds a "relativistic correction" to $\Omega^2 = GM/r^3 \rightarrow$ 'C_SPURIOUS_CORRECTION'.

Full Raw Model Response

****Derivation****

- **Characteristic radii**** (units of GM/c^2): event horizon $2GM/c^2$, photon sphere $3GM/c^2$, ISCO $6GM/c^2$ — three distinct radii. With $GM/c^2 = G \cdot 10 M_\odot / c^2 \approx 1476.6$ km: horizon 2953.2 km, photon sphere 4429.8 km, $r_{\text{ISCO}} = 6 \cdot 1476.6 \approx 8859.6$ km.
- **Orbital frequency.**** $f = \Omega/2\pi = (1/2\pi)\sqrt{(GM/r_{\text{ISCO}}^3)}$, $GM = 1.4766 \times 10^{27} \text{ m}^3/\text{s}^2$, $r_{\text{ISCO}} = 8859.6 \times 10^3 \text{ m} \Rightarrow f_{\text{orb}} = 1.554 \times 10^{-3} \text{ Hz}$.
- **GW frequency.**** Dominant emission at twice the orbital frequency: $f_{\text{GW}} = 2 \cdot 1.554 \times 10^{-3} = 3.108 \times 10^{-3} \text{ Hz}$.
- **Interpretation.**** No stable circular orbit exists inside r_{ISCO} , so the inspiral transitions to plunge/merger there — the quasi-periodic chirp "shuts off." Since $r_{\text{ISCO}} \propto M$ and $f \propto M^{-1}$, heavier binaries merge at lower frequency ($f_{\text{ISCO}} \propto 1/M$).

****FINAL ANSWER****

```
\[
\begin{aligned}
\mathrm{f_{gw}\_hz} & \quad \&= 3.108 \times 10^{-3} \\
\mathrm{f_{orbital}\_hz} & \quad \&= 1.554 \times 10^{-3} \\
\mathrm{r_{isco}\_km} & \quad \&= 8859.6
\end{aligned}
\]
```

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	7.2337	FAIL	6.07229
g_factor	2.002323	2.00014	PASS	-0.002183

Suggested error labels: anomaly_times_1e3:MISMATCH; numeric_mismatch

Failed fields: anomaly_times_1e3

Run ID: b29759a2-66b7-4fc4-ba87-3dfd71ad6ba3

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &\approx 1.16141 \\ \mathrm{g_factor} &\approx 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Schwinger one-loop.**** $a_e = \alpha/(2\pi)$.

2. ****Anomaly.**** $a_e = (1/137.036)/(2\pi) = 0.000072337/____ = 7.2337 \times 10^{-5} \times 10^{-3} = a_e \times 10^3 = 7.2337$.

3. ****g-factor.**** $g = 2(1 + a_e) = 2 \cdot 1.000072337 = 2.000144674 \approx 2.00014$.

4. ****Meaning.**** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{anomaly_times_1e3} &\approx 7.2337 \\ \mathrm{g_factor} &\approx 2.00014 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	1.161	PASS	-0.00041
g_factor	2.002323	2.002323	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: 25529e5b-327b-4589-a52a-93f73ca80330

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Schwinger one-loop.**** $a_e = \frac{\alpha}{2\pi}$.

2. ****Anomaly.**** $a_e = \frac{1/137.036}{2\pi} = \frac{1}{274.072\pi} \approx 0.0011614 \times 10^{-3} \implies \mathbf{a_e \times 10^3 = 1.161}$.

3. ****g-factor.**** $g = 2(1 + a_e) = 2(1 + 0.0011614) = 2(1.0011614) = 2.0023228 \approx \mathbf{2.002323}$.

4. ****Meaning.**** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.161 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	1.16141	PASS	0
g_factor	2.002323	2.002323	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: c4603976-d5d0-4988-a94c-66b6032bc539

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Schwinger one-loop.**** $a_e = \alpha/(2\pi)$.

2. ****Anomaly.**** $a_e = (1/137.036)/(2\pi) = 1/861.022 = 0.00116141 \times 10^{-3} \implies \mathbf{a_e \times 10^3 = 1.16141}$.

3. ****g-factor.**** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. ****Meaning.**** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	1.1614	PASS	-1e-05
g_factor	2.002323	2.002323	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: 4013320a-823c-4eea-a0f8-eff99460982d

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

Derivation

1. **Schwinger one-loop.**
The leading quantum-electrodynamics (QED) correction to the electron's magnetic moment (the anomalous magnetic moment) is given by the Schwinger result:
$$a_e \equiv \frac{g-2}{2} = \frac{\alpha}{2\pi}.$$

2. **Anomaly.**
Insert the given fine-structure constant $(\alpha = 1/137.036)$:
$$a_e = \frac{1/137.036}{2\pi} = \frac{1}{137.036 \times 2\pi} \approx \frac{1}{860.44} \approx 0.0011614.$$

To express this in units of (10^{-3}) :
$$a_e \times 10^3 \approx 1.1614.$$

Hence $a_e \times 10^3 = 1.1614$ (rounded to four decimal places).

3. **g-factor.**
The relation between the anomalous magnetic moment and the g -factor is $(g = 2(1 + a_e))$. Using the value above:
$$g = 2 \bigl(1 + 0.0011614\bigr) = 2 \times 1.0011614 \approx 2.0023228.$$

Rounded to six decimal places, $g_{\text{factor}} = 2.002323$.

4. **Meaning.**
The anomaly (a_e) quantifies the deviation of the electron's magnetic moment from the tree-level Dirac prediction $(g = 2)$. At tree level, a Dirac fermion has exactly $(g = 2)$. The one-loop (Schwinger) correction arises from the electron emitting and reabsorbing a virtual photon, which modifies the magnetic moment. Because this leading correction depends only on the electromagnetic coupling (α) and the geometry of the loop (i.e., $(\alpha/(2\pi))$), it is **mass-independent**. Consequently, the same coefficient applies to the muon at leading order; the muon's larger mass contributes only to higher-order (loop-suppressed) terms, not to the leading $(\alpha/(2\pi))$ piece.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.1614 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	0.1514	FAIL	-1.01001
g_factor	2.002323	2.0003028	PASS	-0.0020202

Suggested error labels: anomaly_times_1e3:MISMATCH; numeric_mismatch

Failed fields: anomaly_times_1e3

Run ID: eaff2b64-dfb3-49b3-9728-2a8a928c5ac7

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

```
## Step 1: **Schwinger one-loop.** a_e = \alpha/(2\pi).
Given \alpha = 1/137.036, we can substitute this value into the formula for a_e.

## Step 2: **Anomaly.** a_e = (1/137.036)/(2\pi) = ____/____ = ____\times 10^{-3} \Rightarrow
**a_e \times 10^3 = ____**
To find a_e, we need to calculate (1/137.036)/(2\pi).

## Step 3: **g-factor.** g = 2(1 + a_e) = 2 \cdot ____ = **____**
We will use the calculated value of a_e to find the g-factor.

## Step 4: **Meaning.** The anomaly is the deviation from the tree-level Dirac value g = 2, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient \alpha/(2\pi) is **mass-independent**, so the muon receives the same \alpha/(2\pi) at leading order; mass-dependent contributions enter only at higher order.

## Step 5: Calculate a_e
a_e = (1/137.036)/(2\pi) = 0.0001514

## Step 6: Calculate a_e \times 10^3
a_e \times 10^3 = 0.1514

## Step 7: Calculate g-factor
g = 2(1 + a_e) = 2(1 + 0.0001514) = 2.0003028

## Step 8: Finalize the answer
anomaly_times_1e3 = 0.1514
g_factor = 2.0003028

The final answer is: \boxed{\begin{aligned} \mathrm{anomaly\_times\_1e3} &= 0.1514 \\ \mathrm{g\_factor} &= 2.0003028 \end{aligned}}
```

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	0.11598805	FAIL	-1.04542
g_factor	2.002323	2.002320	PASS	-3e-06

Suggested error labels: anomaly_times_1e3:MISMATCH; numeric_mismatch

Failed fields: anomaly_times_1e3

Run ID: b3e81597-bbd0-4808-a7b4-5c7c68c104f0

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Schwinger one-loop.**** $(a_e = \alpha/(2\pi))$.

2. ****Anomaly.****
 $(a_e = \frac{1}{137.036} \cdot \frac{1}{2\pi}) = \frac{0.0072973525693}{6.283185307}$
 $\approx 0.0011598805 \times 10^{-3}$
 $\Rightarrow (a_e \times 10^3 = 0.11598805)$.

3. ****g-factor.****
 $(g = 2(1 + a_e) = 2 \cdot (1 + 0.0011598805) = 2.002319761)$
 $\Rightarrow (g = 2.002320)$.

4. ****Meaning.**** The anomaly is the deviation from the tree-level Dirac value $(g = 2)$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $(\alpha/(2\pi))$ is **mass-independent**, so the muon receives the same $(\alpha/(2\pi))$ at leading order; mass-dependent contributions enter only at higher order.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 0.11598805 \\ \mathrm{g_factor} &= 2.002320 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	3.626	FAIL	2.46459
g_factor	2.002323	2.007252	PASS	0.004929

Suggested error labels: anomaly_times_1e3:MISMATCH; numeric_mismatch

Failed fields: anomaly_times_1e3

Run ID: 96198746-3f2d-4926-bcef-e8433bbb98d7

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Schwinger one-loop.**** $(a_e = \frac{\alpha}{2\pi})$.

2. ****Anomaly.**** $(a_e = \frac{1}{137.036} \cdot \frac{1}{2\pi} = \frac{1}{274.072\pi} = 0.003626 \times 10^{-3} \implies \mathbf{a_e \times 10^3 = 3.626})$.

3. ****g-factor.**** $(g = 2(1 + a_e) = 2 \cdot 1.003626 = \mathbf{2.007252})$.

4. ****Meaning.**** The anomaly is the deviation from the tree-level Dirac value $(g = 2)$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $(\frac{\alpha}{2\pi})$ is mass-independent, so the muon receives the same $(\frac{\alpha}{2\pi})$ at leading order; mass-dependent contributions enter only at higher order.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{anomaly_times_1e3} &= 3.626 \\ \mathrm{g_factor} &= 2.007252 \end{aligned}$$

Question

Compute the leading QED correction to the electron's magnetic moment. Use $\alpha = 1/137.036$.

Conventions: anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ (i.e. in units of 10^{-3}) and the g -factor itself.

Tasks:
1. State the one-loop (Schwinger) result for a_e in terms of α .
2. Evaluate $a_e \times 10^3$.
3. Evaluate the g -factor to 6 decimal places.
4. Explain physically what the anomaly measures (tree-level Dirac vs loop corrections), and why the SAME leading coefficient applies to the muon.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
anomaly_times_1e3	1.16141	0.3648675	FAIL	-0.796543
g_factor	2.002323	2.0000003649	PASS	-0.00232264

Suggested error labels: anomaly_times_1e3:MISMATCH; numeric_mismatch

Failed fields: anomaly_times_1e3

Run ID: ca000060-3995-4bae-816b-f16738e1b5cc

Worked Gold Answer

QFT_G2_001 — Electron g-2: the Schwinger term

Given: $\alpha = 1/137.036$. Anomaly $a_e = (g-2)/2$; report $a_e \times 10^3$ and g .

Derivation

1. **Schwinger one-loop.** $a_e = \alpha/(2\pi)$.

2. **Anomaly.** $a_e = (1/137.036)/(2\pi) = 0.0072974/6.2832 = 1.16141 \times 10^{-3} \Rightarrow \mathbf{a_e \times 10^3 = 1.16141}$.

3. **g-factor.** $g = 2(1 + a_e) = 2 \cdot 1.00116141 = \mathbf{2.002323}$.

4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value $g = 2$, produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient $\alpha/(2\pi)$ is **mass-independent**, so the muon receives the same $\alpha/(2\pi)$ at leading order; mass-dependent contributions enter only at higher order.

FINAL ANSWER

$$\begin{aligned} \mathrm{anomaly} \times 10^3 &= 1.16141 \\ \mathrm{g_factor} &= 2.002323 \end{aligned}$$

Reasoning checkpoints: $a_e = \alpha/(2\pi)$ (Schwinger); $a_e \times 10^3 \approx 1.1614$, $g \approx 2.002323$; anomaly = loop correction to Dirac $g = 2$; leading coefficient mass-independent (same $\alpha/2\pi$ for the muon); does not report α/π or confuse a_e with $g - 2$.

Wrong-answer fingerprints: uses α/π (factor 2) $\rightarrow$ 'N_2PI_VS_PI'; conflates a_e with $g - 2 \rightarrow$ 'C_ANOMALY_G_CONFLATED'.

Full Raw Model Response

```
**Derivation**
1. **Schwinger one-loop.**  $a_e = \alpha/(2\pi)$ .
2. **Anomaly.**  $a_e = (1/137.036)/(2\pi) = 0.000729735/2 = 0.0003648675 \times 10^{-3} \Rightarrow a_e \times 10^3 = 0.3648675$ .
3. **g-factor.**  $g = 2(1 + a_e) = 2 \cdot 1.000003648675 = 2.000003649$ .
4. **Meaning.** The anomaly is the deviation from the tree-level Dirac value  $g = 2$ , produced by the electron emitting and reabsorbing a virtual photon (one loop). The leading coefficient  $\alpha/(2\pi)$  is mass-independent, so the muon receives the same  $\alpha/(2\pi)$  at leading order; mass-dependent contributions enter only at higher order.
**FINAL ANSWER**

\begin{aligned} \mathrm{anomaly} \times 10^3 &= 0.3648675 \\ \mathrm{g\_factor} &= 2.000003649 \end{aligned}
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.933	PASS	-0.000333
classical_limit	0.666667	0.667	PASS	0.000333
f_threshold	0.5	0.5	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: 30ac6932-dc57-4cf3-bf98-468533408eb0

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

****Derivation****

- **Output fidelity.**** $F = (2 \cdot 0.900 + 1)/3 = 2.8/3 = \mathbf{0.933}$.
- **Classical limit.**** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.667}$ (Massar–Popescu).
- **Threshold to beat it.**** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- **Interpretation.**** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.933$.

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{classical\_limit} &= 0.667 \\\mathrm{f\_threshold} &= 0.5 \\\mathrm{teleport\_fidelity} &= 0.933 \\\end{aligned}\]
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.933	PASS	-0.000333
classical_limit	0.666667	0.667	PASS	0.000333
f_threshold	0.5	0.5	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: 7c096d8f-6286-4837-bb5c-38e6df0e6b56

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies \mathbf{f > 1/2 = 0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = 2.8/3 = 0.933$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $F_{cl} = 2/3 = 0.667$  (Massar–Popescu).
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = 0.5$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = 0.933$ .
**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= 0.667 \\\mathrm{f\_threshold} &= 0.5 \\\mathrm{teleport\_fidelity} &= 0.933 \end{aligned}\]
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.933	PASS	-0.000333
classical_limit	0.666667	0.667	PASS	0.000333
f_threshold	0.5	0.500	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: 42445057-c8f5-4253-80d2-5a671b2f970e

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933}$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $F_{cl} = 2/3 = \mathbf{0.667}$  (Massar–Popescu).
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.500}$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = 0.933$ .

**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= \mathbf{0.667} \\\mathrm{f\_threshold} &= \mathbf{0.500} \\\mathrm{teleport\_fidelity} &= \mathbf{0.933} \\\end{aligned}\]
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	$\frac{2 \times 0.900 + 1}{3} \approx 0.9333$	PASS	
classical_limit	0.666667	$\frac{2}{3}$	PASS	
f_threshold	0.5	$\frac{1}{2}$	PASS	

Suggested error labels: none

Failed fields: none

Run ID: b82bd97e-a127-4a93-adc1-3fd1abd73acc

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F = (2f+1)/3 = 0.9333$; classical limit $2/3$; beats classical iff $f > 1/2$; $f > 1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**

1. **Output fidelity.**
  \[F = \frac{2f + 1}{3} = \frac{2 \times 0.900 + 1}{3} = \frac{1.8 + 1}{3} = \frac{2.8}{3} \approx 0.9333\].
  So,  $F \approx 0.933$ .

2. **Classical limit.**
  The best average fidelity achievable with a measure-and-send protocol (no entanglement) is the Massar–Popescu limit:
  \[F_{\text{cl}} = \frac{2}{3} \approx 0.6667\].
  Hence,  $F_{cl} = 2/3$ .

3. **Threshold to beat it.**
  Require  $(\frac{2f + 1}{3} > \frac{2}{3})$ .
  Multiply by 3:  $(2f + 1 > 2 \implies 2f > 1 \implies f > 1/2)$ .
  Therefore, the minimum singlet fidelity needed to surpass the classical limit is  $f > 0.5$ .

4. **Interpretation.**
  For an isotropic noise channel, the condition  $(f > 1/2)$  precisely marks the point where the shared state possesses non-zero entanglement. Since the classical (measure-and-send) strategy cannot exceed a fidelity of  $(2/3)$ , achieving any teleportation fidelity  $(F > 2/3)$  (as we do with  $(F \approx 0.933)$ ) demonstrates that the channel is genuinely quantum and benefits from entanglement assistance. Note that the given singlet fidelity  $(f = 0.900)$  yields a teleportation fidelity  $(F \approx 0.933)$ , which is indeed higher than the classical limit.

**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= \frac{2}{3} \\\mathrm{f\_threshold} &= \frac{1}{2} \\\mathrm{teleport\_fidelity} &= \frac{2 \times 0.900 + 1}{3} \approx 0.933\end{aligned}\]
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.9333	PASS	-3.3e-05
classical_limit	0.666667	0.6667	PASS	3.3e-05
f_threshold	0.5	0.5	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: ba60237f-f38f-49a1-aff7-f714152bab19

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{\text{cl}} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

$$\begin{aligned} & \mathrm{teleport_fidelity} \quad \&= \quad 0.933333 \\ & \mathrm{classical_limit} \quad \&= \quad 0.666667 \\ & \mathrm{f_threshold} \quad \&= \quad 0.5 \end{aligned}$$

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = (1.8 + 1)/3 = 2.8/3 = 0.9333$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $F_{\text{cl}} = 2/3 = 0.6667$  (Massar–Popescu).
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = 0.5$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = 0.9333$ .

**FINAL ANSWER**
\begin{aligned}
&\mathrm{classical\_limit} \quad \&= \quad 0.6667 \\
&\mathrm{f\_threshold} \quad \&= \quad 0.5 \\
&\mathrm{teleport\_fidelity} \quad \&= \quad 0.9333
\end{aligned}
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	$\frac{2.8}{3}$	PASS	
classical_limit	0.666667	$0.666\ldots$	FAIL	
f_threshold	0.5	0.5	PASS	0

Suggested error labels: classical_limit:UNPARSEABLE:parse-fail:TokenError; unparseable_value

Failed fields: classical_limit

Run ID: 9584a665-3773-4c3a-a6fd-e7fe2777cbbb

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900$ ($= f$) $\rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = 2.8/3 = **2.8/3**$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $**F_{cl} = 2/3 = 0.666\ldots**$  (Massar–Popescu).
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies **f > 1/2 = 0.5**$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = **2.8/3**$ .

**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= 0.666\ldots \\\mathrm{f\_threshold} &= 0.5 \\\mathrm{teleport\_fidelity} &= \frac{2.8}{3}\end{aligned}\]
```


Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi^{\sim} \rangle$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.6	FAIL	-0.333333
classical_limit	0.666667	0.666...	PASS	
f_threshold	0.5	0.5	PASS	0

Suggested error labels: teleport_fidelity:MISMATCH; numeric_mismatch

Failed fields: teleport_fidelity

Run ID: ef9cc4a7-79d5-4db1-94dc-31e32778b632

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \Rightarrow 2f+1 > 2 \Rightarrow \mathbf{f > 1/2 = 0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = 1.8/3 = \mathbf{0.6}$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $F_{cl} = 2/3 = 0.666\dots$ .
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \Rightarrow 2f+1 > 2 \Rightarrow \mathbf{f > 1/2 = 0.5}$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = 0.6$ .
**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= 0.666\dots \\\mathrm{f\_threshold} &= 0.5 \\\mathrm{teleport\_fidelity} &= 0.6\end{aligned}\]
```

Question

Standard teleportation is performed with a noisy shared resource of singlet fidelity $f = \langle \chi | \Psi \rangle = 0.900$ (isotropic noise). Inputs are uniformly random pure qubits.

Conventions: average teleportation fidelity for an isotropic resource is $F = (2f+1)/3$; 'classical limit' = best average fidelity with measure-and-send and no entanglement.

Tasks:

- Evaluate the average teleportation fidelity F for $f = 0.900$.
- State and evaluate the classical limit.
- What minimum singlet fidelity f beats the classical limit?
- Explain why $f > 1/2$ relates to entanglement of the isotropic resource, and why beating $2/3$ certifies a quantum channel.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
teleport_fidelity	0.933333	0.933	PASS	-0.000333
classical_limit	0.666667	0.666	PASS	-0.000667
f_threshold	0.5	0.500	PASS	0

Suggested error labels: none

Failed fields: none

Run ID: dfc152eb-9100-4902-bb40-b122e3842d5f

Worked Gold Answer

QI_TELEPORT_001 — Teleportation with a noisy resource

Given: singlet fidelity $f = \langle \Psi | \chi | \Psi \rangle = 0.900$ (isotropic). Average teleportation fidelity $F = (2f+1)/3$.

Derivation

- Output fidelity.** $F = (2 \cdot 0.900 + 1)/3 = 2.800/3 = \mathbf{0.933333}$.
- Classical limit.** Best measure-and-send strategy with no shared entanglement: $F_{cl} = 2/3 = \mathbf{0.666667}$ (Massar–Popescu).
- Threshold to beat it.** $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = \mathbf{0.5}$.
- Interpretation.** $f > 1/2$ is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds $2/3$, any $F > 2/3$ certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality $f = 0.900$ is **not** the teleportation fidelity $F = 0.9333$.

FINAL ANSWER

```
\[\begin{aligned}\mathrm{teleport\_fidelity} &= 0.933333 \\\mathrm{classical\_limit} &= 0.666667 \\\mathrm{f\_threshold} &= 0.5\end{aligned}\]
```

Reasoning checkpoints: $F=(2f+1)/3=0.9333$; classical limit $2/3$; beats classical iff $f>1/2$; $f>1/2$ is the isotropic-state entanglement threshold; does not report f itself as F .

Wrong-answer fingerprints: reports $F = 0.900 (= f) \rightarrow$ 'C_RESOURCE_OUTPUT_CONFLATED'; classical limit $\neq 2/3 \rightarrow$ 'N_CLASSICAL_LIMIT_WRONG'.

Full Raw Model Response

```
**Derivation**
1. **Output fidelity.**  $F = (2 \cdot 0.900 + 1)/3 = 2.8/3 = \mathbf{0.933}$ .
2. **Classical limit.** Best measure-and-send strategy with no shared entanglement:  $F_{cl} = 2/3 = 0.666$  (Massar–Popescu).
3. **Threshold to beat it.**  $(2f+1)/3 > 2/3 \implies 2f+1 > 2 \implies f > 1/2 = 0.500$ .
4. **Interpretation.**  $f > 1/2$  is exactly the entanglement threshold of the isotropic resource; since no classical strategy exceeds  $2/3$ , any  $F > 2/3$  certifies a genuinely quantum (entanglement-assisted) channel. Note the bookkeeping: the resource quality  $f = 0.900$  is not the teleportation fidelity  $F = 0.933$ .
**FINAL ANSWER**

\[\begin{aligned}\mathrm{classical\_limit} &= 0.666 \\\mathrm{f\_threshold} &= 0.500 \\\mathrm{teleport\_fidelity} &= 0.933 \\\end{aligned}\]
```

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:
1. Compute the solid angle Ω subtended by the cone at the origin.
2. State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
3. How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
4. Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	3.1416	PASS	7e-06
berry_phase_rad	1.570796	1.5708	PASS	4e-06

Suggested error labels: none

Failed fields: none

Run ID: d94b3e8b-d2fa-4bfd-8261-18fc32f1e6c2

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

1. Solid angle. $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.141593 \text{ sr}$.

2. Berry phase (spin-½). $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.570796 \text{ rad}$. The factor ½ is the spin.

3. Spin-1 (m = 1). In general $|\gamma| = m\Omega$ for the aligned |S,m> state, so spin-1 m = 1 gives $|\gamma| = \pi$ (scales with m).

4. Why geometric. γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{solid_angle_sr} &= 3.141593 \\ \mathrm{berry_phase_rad} &= 1.570796 \end{aligned}$$

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) → 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase → 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

Derivation

1. **Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi \text{ sr}$.

2. **Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 \text{ rad}$. The factor ½ is the spin.

3. **Spin-1 (m = 1).** In general $|\gamma| = m\Omega$ for the aligned |S,m> state, so spin-1 m = 1 gives $|\gamma| = \pi$ (scales with m).

4. **Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{berry_phase_rad} &= 1.5708 \\ \mathrm{solid_angle_sr} &= 3.1416 \end{aligned}$$

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:
1. Compute the solid angle Ω subtended by the cone at the origin.
2. State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
3. How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
4. Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	3.14159	PASS	-3e-06
berry_phase_rad	1.570796	1.5708	PASS	4e-06

Suggested error labels: none

Failed fields: none

Run ID: 5adb68d3-d803-46a1-ac23-424753a795f6

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

1. Solid angle. $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.141593 \text{ sr}$.

2. Berry phase (spin-½). $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.570796 \text{ rad}$. The factor ½ is the spin.

3. Spin-1 (m = 1). In general $|\gamma| = m\cdot\Omega$ for the aligned |S,m> state, so spin-1 m = 1 gives $|\gamma| = \pi$ (scales with m).

4. Why geometric. γ depends only on the solid angle Ω enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{solid_angle_sr} &= 3.141593 \\ \mathrm{berry_phase_rad} &= 1.570796 \end{aligned}$$

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) → 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase → 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

Derivation

1. **Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.14159 \text{ sr}$.

2. **Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.5708 \text{ rad}$. The factor $\frac{1}{2}$ is the spin.

3. **Spin-1 (m = 1).** In general $|\gamma| = m\cdot\Omega$ for the aligned |S,m> state, so spin-1 m = 1 gives $|\gamma| = \pi$ (scales with m).

4. **Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path – not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{berry_phase_rad} &= 1.5708 \\ \mathrm{solid_angle_sr} &= 3.14159 \end{aligned}$$

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:

- Compute the solid angle Ω subtended by the cone at the origin.
- State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
- How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
- Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	π	PASS	
berry_phase_rad	1.570796	$\pi/2$	PASS	

Suggested error labels: none

Failed fields: none

Run ID: 9db36a40-841b-4808-bbed-447ab10fa45e

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

- Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = \mathbf{3.141593\ sr}$.
- Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = \mathbf{1.570796\ rad}$. The factor ½ is the spin.
- Spin-1 ($m = 1$).** In general $|\gamma| = m\cdot\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).
- Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto$ energy $\times$ time).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{solid\_angle\_sr} \quad &= 3.141593 \\\mathrm{berry\_phase\_rad} \quad &= 1.570796\end{aligned}\]
```

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) $\rightarrow$ 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase $\rightarrow$ 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

```
**Derivation**
1. **Solid angle.**  $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi$  sr.
2. **Berry phase (spin-½).**  $\gamma = -\frac{1}{2}\Omega$ , so  $|\gamma| = \Omega/2 = \pi/2 = 1.571$  rad. The factor ½ is the spin.
3. **Spin-1 ( $m = 1$ ).|\gamma| = m\cdot\Omega for the aligned  $|S,m\rangle$  state, so spin-1  $m = 1$  gives  $|\gamma| = \pi$  (scales with  $m$ ).
4. **Why geometric.**  $\gamma$  depends only on the solid angle  $\Omega$  enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ( $\propto$  energy  $\times$  time).

**FINAL ANSWER**

\[\begin{aligned}\mathrm{solid\_angle\_sr} \quad &= \pi \\\mathrm{berry\_phase\_rad} \quad &= \pi/2\end{aligned}\]
```

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:

- Compute the solid angle Ω subtended by the cone at the origin.
- State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
- How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
- Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	π	PASS	
berry_phase_rad	1.570796	$\frac{\pi}{2}$	PASS	

Suggested error labels: none

Failed fields: none

Run ID: bfead667-22b7-4add-82bd-2984b62eeac1

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report [Berry phase] in radians.

Derivation

- Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = \mathbf{3.141593\ sr}$.
- Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = \mathbf{1.570796\ rad}$. The factor ½ is the spin.
- Spin-1 ($m = 1$).** In general $|\gamma| = m\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).
- Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\sim$ energy $\times$ time).

FINAL ANSWER

$$\begin{aligned} \mathrm{solid_angle_sr} \quad &= 3.141593 \\ \mathrm{berry_phase_rad} \quad &= 1.570796 \end{aligned}$$

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) → 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase → 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

```
**Derivation**

1. **Solid angle.**
\[\Omega = 2\pi\bigl(1 - \cos\theta\bigr) = 2\pi\bigl(1 - \cos 60^\circ\bigr) = 2\pi\bigl(1 - 0.5\bigr) = 2\pi \times 0.5 = \pi\];\text{sr}\]

2. **Berry phase (spin-½).**
For a spin-½ particle the geometric (Berry) phase of the aligned state after one adiabatic cycle is
\[\gamma = -\frac{1}{2}\Omega\]
Hence the magnitude is
\[\gamma] = \frac{\Omega}{2} = \frac{\pi}{2}\];\text{rad}\]

3. **Spin-1 ( $m = 1$ ).**
In general, for a spin- $S$  particle the Berry phase of the aligned  $|S,m\rangle$  state is proportional to the magnetic quantum number  $m$ :
\[\gamma = -m\Omega\]
For spin-1 ( $S=1$ ) with  $m=1$ :
\[\gamma] = \Omega = \pi\]
Wait, that's not right. The correct scaling is  $|\gamma| = m\Omega$ . For  $S=1, m=1$ :
\[\gamma] = \Omega = \pi\]
Actually, the standard result for a spin-1 particle is  $|\gamma| = m\Omega$ . With  $S=1, m=1$ :
\[\gamma] = \Omega = \pi\]
However, many texts state that the Berry phase for a spin- $S$  particle is  $|\gamma| = m\Omega$ . For  $S=1, m=1$  this gives  $|\gamma| = \pi$ . But the problem likely expects the scaling with  $m$ : for spin-1,  $|\gamma| = \Omega$  when  $m=1$ . Let me re-derive: The Berry phase for a spin- $S$  state is  $\gamma = -\frac{m}{S}\Omega$ . For  $S=1, m=1$ :  $\gamma = -\Omega$ . So the magnitude is  $|\gamma| = \Omega$ .
**Correction:** The magnitude for spin-1,  $m=1$ , is  $|\gamma| = \Omega = \pi$  rad, not  $|\gamma| = \pi/2$ . The scaling is  $|\gamma| = m\Omega$ .
Therefore, for spin-1,  $m=1$ :
\[\gamma] = \Omega = \pi\];\text{rad}\]

4. **Why geometric.**
The Berry phase depends solely on the solid angle  $\Omega$  enclosed by the path of the magnetic field direction in parameter space. It is independent of:
- the speed at which the field is traversed (as long as the evolution remains adiabatic),
- the magnitude of the magnetic field, and
- the total elapsed time of the cycle.
It is a purely geometric property, distinct from the dynamical phase, which depends on the energy and the duration of the evolution.

**FINAL ANSWER**

\[\begin{aligned} \mathrm{berry\_phase\_rad} \quad &= \pi \\ \mathrm{solid\_angle\_sr} \quad &= \pi \end{aligned}\]
```

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:

- Compute the solid angle Ω subtended by the cone at the origin.
- State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
- How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
- Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	3.14159	PASS	-3e-06
berry_phase_rad	1.570796	1.5708	PASS	4e-06

Suggested error labels: none

Failed fields: none

Run ID: e85fdb3c-6f96-43c1-ae20-f3af6c23ab26

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

- Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = \mathbf{3.141593\text{ sr}}$.
- Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = \mathbf{1.570796\text{ rad}}$. The factor ½ is the spin.
- Spin-1 ($m = 1$).** In general $|\gamma| = m\cdot\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).
- Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto$ energy $\times$ time).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{solid\_angle\_sr} \quad &= 3.141593 \\\mathrm{berry\_phase\_rad} \quad &= 1.570796\end{aligned}\]
```

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) $\rightarrow$ 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase $\rightarrow$ 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

****Derivation****

1. ****Solid angle.**** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = 2\pi(0.5) = \pi = 3.14159\text{ sr}$.

2. ****Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.5708\text{ rad}$. The factor ½ is the spin.

3. ****Spin-1 ($m = 1$).** In general $|\gamma| = m\cdot\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).

4. ****Why geometric.**** γ depends only on the solid angle Ω enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto$ energy $\times$ time).

****FINAL ANSWER****

```
\[\begin{aligned}\mathrm{berry\_phase\_rad} \quad &= 1.5708 \\\mathrm{solid\_angle\_sr} \quad &= 3.14159\end{aligned}\]
```

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:

- Compute the solid angle Ω subtended by the cone at the origin.
- State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
- How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
- Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	π	PASS	
berry_phase_rad	1.570796	$\pi/2$	PASS	

Suggested error labels: none

Failed fields: none

Run ID: fdc95d2c-3de5-4854-8c60-7b922434ac4c

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

- Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = \mathbf{3.141593\ sr}$.
- Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = \mathbf{1.570796\ rad}$. The factor ½ is the spin.
- Spin-1 ($m = 1$).** In general $|\gamma| = m\cdot\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).
- Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto$ energy $\times$ time).

FINAL ANSWER

```
\[\begin{aligned}\mathrm{solid\_angle\_sr} \quad &= 3.141593 \\\mathrm{berry\_phase\_rad} \quad &= 1.570796\end{aligned}\]
```

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) $\rightarrow$ 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase $\rightarrow$ 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

```
**Derivation**
1. **Solid angle.**  $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = \pi\ sr$ .
2. **Berry phase (spin-½).**  $\gamma = -\frac{1}{2}\Omega$ , so  $|\gamma| = \Omega/2 = \pi/2 = \pi/2\ rad$ . The factor ½ is the spin.
3. **Spin-1 ( $m = 1$ ).|\gamma| = m\cdot\Omega for the aligned  $|S,m\rangle$  state, so spin-1  $m = 1$  gives  $|\gamma| = \pi$  (scales with  $m$ ).
4. **Why geometric.**  $\gamma$  depends only on the solid angle  $\Omega$  enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ( $\propto$  energy  $\times$  time).

**FINAL ANSWER**

\[\begin{aligned}\mathrm{berry\_phase\_rad} \quad &= \pi/2 \\\mathrm{solid\_angle\_sr} \quad &= \pi\end{aligned}\]
```


Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:
1. Compute the solid angle Ω subtended by the cone at the origin.
2. State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
3. How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
4. Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
solid_angle_sr	3.141593	π	PASS	
berry_phase_rad	1.570796	$\frac{\pi}{2}$	PASS	

Suggested error labels: none

Failed fields: none

Run ID: 183a2bc8-11a9-43e6-a8a0-67b79cff192b

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

1. Solid angle. $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.141593 \text{ sr}$.

2. Berry phase (spin-½). $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.570796 \text{ rad}$. The factor ½ is the spin.

3. Spin-1 ($m = 1$). In general $|\gamma| = m\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).

4. Why geometric. γ depends only on the solid angle Ω enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{solid_angle_sr} &= 3.141593 \\ \mathrm{berry_phase_rad} &= 1.570796 \end{aligned}$$

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) → 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase → 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

Derivation

1. **Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi \text{ sr}$.

2. **Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = \pi/2 \text{ rad}$. The factor ½ is the spin.

3. **Spin-1 ($m = 1$).|\gamma| = m\Omega for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).

4. **Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — not on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{berry_phase_rad} &= \frac{\pi}{2} \\ \mathrm{solid_angle_sr} &= \pi \end{aligned}$$

Question

A spin-1/2 particle is in the instantaneous aligned state of a magnetic field $\vec{B}(t)$ whose direction traces, adiabatically and once, a cone of half-angle $\theta = 60^\circ$ about the z -axis.

Conventions: 'Berry phase' means the geometric phase of the adiabatically transported aligned state after one full cycle; report its magnitude in radians.

Tasks:
1. Compute the solid angle Ω subtended by the cone at the origin.
2. State the Berry phase for a spin-1/2 aligned state in terms of Ω , and evaluate its magnitude.
3. How would the result change for total spin $S = 1$ (aligned, $m = 1$)?
4. Explain why this phase is geometric: what does it NOT depend on?

Numeric Comparison

field_name	expected	model_value	match	delta_or_sta
solid_angle_sr	3.141593	3.141592653589793	PASS	-3.4641e-07
berry_phase_rad	1.570796	1.5707963267948966	PASS	3.26795e-07

Suggested error labels: none

Failed fields: none

Run ID: 0c429cf4-5519-4352-b10b-b78367d1aa96

Worked Gold Answer

QM_BERRY_001 — Berry phase: spin-½ on a cone

Given: spin-½ in the instantaneous aligned state; field direction traces a cone of half-angle $\theta = 60^\circ$ once, adiabatically. Report |Berry phase| in radians.

Derivation

1. Solid angle. $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.141593 \text{ sr}$.

2. Berry phase (spin-½). $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.570796 \text{ rad}$. The factor ½ is the spin.

3. Spin-1 ($m = 1$). In general $|\gamma| = m\Omega$ for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).

4. Why geometric. γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{solid_angle_sr} &= 3.141593 \\ \mathrm{berry_phase_rad} &= 1.570796 \end{aligned}$$

Reasoning checkpoints: $\Omega = 2\pi(1 - \cos 60^\circ) = \pi$; spin-½ $|\gamma| = \Omega/2 = \pi/2$ (factor ½ present); spin-1 $|\gamma| = m\Omega = \pi$; geometric (independent of speed, distinct from dynamical); does not report Ω itself as the phase.

Wrong-answer fingerprints: reports $\Omega = \pi$ as the phase (½ dropped) → 'N_FACTOR_HALF_DROPPED'; conflates dynamical and geometric phase → 'C_DYNAMICAL_GEOMETRIC_CONFLATED'.

Full Raw Model Response

Derivation

1. **Solid angle.** $\Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \cos 60^\circ) = 2\pi(1 - 0.5) = \pi = 3.141592653589793 \text{ sr}$.

2. **Berry phase (spin-½).** $\gamma = -\frac{1}{2}\Omega$, so $|\gamma| = \Omega/2 = \pi/2 = 1.5707963267948966 \text{ rad}$. The factor ½ is the spin.

3. **Spin-1 ($m = 1$).|\gamma| = m\Omega for the aligned $|S,m\rangle$ state, so spin-1 $m = 1$ gives $|\gamma| = \pi$ (scales with m).

4. **Why geometric.** γ depends only on the solid angle Ω enclosed by the field's path — **not** on the traversal speed (as long as adiabatic), the field magnitude, or elapsed time. It is distinct from the dynamical phase ($\propto \text{energy} \times \text{time}$).

FINAL ANSWER

$$\begin{aligned} \mathrm{berry_phase_rad} &= 1.5707963267948966 \\ \mathrm{solid_angle_sr} &= 3.141592653589793 \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.267	FAIL	0.079955
distance_ly	0.563194	1.62	FAIL	1.05681
final_beta	0.774547	0.752	FAIL	-0.022547

Suggested error labels:
distance_ly:MISMATCH;final_beta:MISMATCH;lab_time_yr:MISMATCH;numeric_mismatch

Failed fields: distance_ly;final_beta;lab_time_yr

Run ID: 0fdb6f7-aa2f-4d13-85e3-1de67d52d0f1

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation

1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER

$$\begin{aligned} \backslash \backslash \\ \backslash \mathrm{lab_time_yr} & \quad \&= \quad 1.187045 \quad \backslash \backslash \\ \backslash \mathrm{distance_ly} & \quad \&= \quad 0.563194 \quad \backslash \backslash \\ \backslash \mathrm{final_beta} & \quad \&= \quad 0.774547 \\ \backslash \mathrm{end{aligned}} \\ \backslash \end{aligned}$$

Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly ; $\beta \approx 0.77$; rejects $v = a\tau$ — rapidity grows linearly, not velocity.

Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****
1. ****Exact hyperbolic motion.**** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the ****rapidity****.
2. ****Rapidity & lab time.**** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0327$. Lab time $= (c/a) \sinh(1.0327) \Rightarrow$ ****1.267 yr**** (> 1 proper year — the correct dilation direction).
3. ****Distance.**** $x = (c^2/a)(\cosh(1.0327) - 1) \Rightarrow$ ****1.62 ly****.
4. ****Final speed.**** $\beta = \tanh(1.0327) =$ ****0.752****.
5. ****The student's error.**** $v = a\tau$ gives $\beta = a\tau/c = 1.0327 > 1$ (superluminal). The resolution: the ****rapidity**** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.
****FINAL ANSWER****

$$\begin{aligned} \backslash \backslash \\ \backslash \mathrm{distance_ly} & \quad \&= \quad 1.62 \quad \backslash \backslash \\ \backslash \mathrm{final_beta} & \quad \&= \quad 0.752 \quad \backslash \backslash \\ \backslash \mathrm{lab_time_yr} & \quad \&= \quad 1.267 \quad \backslash \backslash \\ \backslash \mathrm{end{aligned}} \\ \backslash \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.18	PASS	-0.007045
distance_ly	0.563194	0.56	PASS	-0.003194
final_beta	0.774547	0.775	PASS	0.000453

Suggested error labels: none

Failed fields: none

Run ID: 4e6f8a7e-4f83-4d56-8303-b5235d10c856

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation

1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER

$$\begin{aligned} \mathrm{lab_time_yr} &= 1.187045 \\ \mathrm{distance_ly} &= 0.563194 \\ \mathrm{final_beta} &= 0.774547 \end{aligned}$$

Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly; β 0.77; rejects $v = a\tau$ — rapidity grows linearly, not velocity.

Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Exact hyperbolic motion.**** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the ****rapidity****.

2. ****Rapidity & lab time.**** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0326$. Lab time $= (c/a) \sinh(1.0326) \Rightarrow$ ****1.18 yr**** (> 1 proper year — the correct dilation direction).

3. ****Distance.**** $x = (c^2/a)(\cosh(1.0326) - 1) \Rightarrow$ ****0.56 ly****.

4. ****Final speed.**** $\beta = \tanh(1.0326) =$ ****0.775****.

5. ****The student's error.**** $v = a\tau$ gives $\beta = a\tau/c = 1.0326 > 1$ (superluminal). The resolution: the ****rapidity**** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{distance_ly} &= 0.56 \\ \mathrm{final_beta} &= 0.775 \\ \mathrm{lab_time_yr} &= 1.18 \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.47	FAIL	0.282955
distance_ly	0.563194	0.97	FAIL	0.406806
final_beta	0.774547	0.775	PASS	0.000453

Suggested error labels:
distance_ly:MISMATCH;lab_time_yr:MISMATCH; numeric_mismatch

Failed fields: distance_ly;lab_time_yr

Run ID: 03445593-5565-4d84-808b-961d8347ef18

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation

1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER

$$\begin{aligned} \mathrm{lab_time_yr} &\approx 1.187045 \\ \mathrm{distance_ly} &\approx 0.563194 \\ \mathrm{final_beta} &\approx 0.774547 \end{aligned}$$

Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly ; $\beta \approx 0.77$; rejects $v = a\tau$ — rapidity grows linearly, not velocity.

Wrong-answer fingerprints: $v = a\tau$ / $\beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Exact hyperbolic motion.****
For constant proper acceleration a , the lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ are given by:
$$\begin{aligned} t(\tau) &= \frac{c}{a} \sinh\left(\frac{a\tau}{c}\right), \\ x(\tau) &= \frac{c^2}{a} (\cosh\left(\frac{a\tau}{c}\right) - 1), \\ \beta(\tau) &= \tanh\left(\frac{a\tau}{c}\right). \end{aligned}$$
Here $\frac{a\tau}{c}$ is the rapidity.

2. ****Rapidity & lab time.****
Compute the rapidity for $\tau = 1$ proper year:
$$\frac{a\tau}{c} = \frac{9.8 \text{ m/s}^2 \times 3.15576 \times 10^7 \text{ s}}{2.99792458 \times 10^8 \text{ m/s}} \approx 1.0316$$
Lab time elapsed:
$$t = \frac{c}{a} \sinh(1.0316) \approx 3.058 \times 10^7 \text{ s} \approx 1.515 \text{ yr}$$
Convert to years:
$$t \approx \frac{3.058 \times 10^7 \text{ s}}{3.15576 \times 10^7 \text{ s/yr}} \approx 1.47 \text{ yr}$$
****Lab time = 1.47 yr**** (greater than the 1 proper year, as expected from time dilation).

3. ****Distance.****
$$x = \frac{c^2}{a} (\cosh(1.0316) - 1) \approx 0.563 \text{ ly}$$
Compute $\cosh(1.0316) \approx 1.666$.
$$x \approx \frac{8.9876 \times 10^{16} \text{ m}}{9.176 \times 10^{15} \text{ m/ly}} \approx 9.69 \text{ ly}$$
Convert to light-years using $1 \text{ ly} = c \times 1 \text{ yr} = 2.9979 \times 10^8 \text{ m}$:
$$\frac{9.69 \times 10^{15} \text{ m}}{2.9979 \times 10^8 \text{ m}} \approx 0.97 \text{ ly}$$
****Distance $\approx 0.97 \text{ ly}$ ****

4. ****Final speed.****
$$\beta = \tanh(1.0316) \approx 0.775$$
****Final speed $\beta \approx 0.775$ **** (sub-luminal, as required).

5. ****The student's error.****
The student used the Newtonian relation $v = a\tau$, giving $\beta = a\tau/c = 1.0316 > 1$, which predicts superluminal speeds. The mistake is treating proper acceleration as if it linearly adds to velocity in the lab frame. In special relativity, velocities combine via rapidities: $\beta = \tanh(\text{rapidity})$, which always stays below 1. Thus the correct velocity never exceeds the speed of light.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{distance_ly} &\approx 0.97 \\ \mathrm{final_beta} &\approx 0.775 \\ \mathrm{lab_time_yr} &\approx 1.47 \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.016	FAIL	-0.171045
distance_ly	0.563194	0.00016	FAIL	-0.563034
final_beta	0.774547	0.032	FAIL	-0.742547

Suggested error labels:
distance_ly:MISMATCH;final_beta:MISMATCH;lab_time_yr:MISMATCH;numeric_mismatch

Failed fields: distance_ly;final_beta;lab_time_yr

Run ID: ea3655fd-8b19-4f34-972d-8d47bb3d3a2f

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation
1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER
$$\begin{aligned} \mathrm{lab_time_yr} &\approx 1.187045 \\ \mathrm{distance_ly} &\approx 0.563194 \\ \mathrm{final_beta} &\approx 0.774547 \end{aligned}$$
Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly; β 0.77; rejects $v = a\tau$ — rapidity grows linearly, not velocity.
Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

```
**Derivation**
1. **Exact hyperbolic motion.**
t(τ) = (c/a) sinh(aτ/c), x(τ) = (c²/a)(cosh(aτ/c) - 1), β(τ) =
tanh(aτ/c). The dimensionless combination aτ/c is the **rapidity**.
2. **Rapidity & lab time.**
aτ/c = 9.8·3.15576×10⁷ / 2.99792458×10⁸ = 0.0325.
Lab time = (c/a) sinh(0.0325) = 1.016 yr.
3. **Distance.**
x = (c²/a)(cosh(0.0325) - 1) = 0.00016 ly.
4. **Final speed.**
β = tanh(0.0325) = 0.032.
5. **The student's error.**
v = aτ gives β = aτ/c = 0.0325 > 1 (superluminal). The resolution: the
**rapidity** aτ/c grows linearly, but velocity β = tanh(rapidity)
saturates below 1. Newtonian kinematics wrongly adds velocities instead
of rapidities.

**FINAL ANSWER**
\[\begin{aligned}
\mathrm{distance\_ly} &\approx 0.00016 \\
\mathrm{final\_beta} &\approx 0.032 \\
\mathrm{lab\_time\_yr} &\approx 1.016
\end{aligned}\]
```


Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.016	FAIL	-0.171045
distance_ly	0.563194	0.521	FAIL	-0.042194
final_beta	0.774547	0.1032	FAIL	-0.671347

Suggested error labels:
distance_ly:MISMATCH;final_beta:MISMATCH;lab_time_yr:MISMATCH;numeric_mismatch

Failed fields: distance_ly;final_beta;lab_time_yr

Run ID: b43a15c6-f44a-490f-9780-8498520180c6

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation

1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER

$$\begin{aligned} \mathrm{lab_time_yr} &= 1.187045 \\ \mathrm{distance_ly} &= 0.563194 \\ \mathrm{final_beta} &= 0.774547 \end{aligned}$$

Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly ; $\beta \approx 0.77$; rejects $v = a\tau$ — rapidity grows linearly, not velocity.

Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Exact hyperbolic motion.****
The dimensionless rapidity is η
$$\eta = \frac{a\tau}{c}.$$

Then
$$\begin{aligned} t(\tau) &= \frac{c}{a} \sinh \eta, \\ x(\tau) &= \frac{c^2}{a} (\cosh \eta - 1), \\ \beta(\tau) &= \tanh \eta. \end{aligned}$$

2. ****Rapidity & lab time.****
Compute η
$$\eta = \frac{9.8 \times 3.15576 \times 10^7}{2.99792458 \times 10^8} \approx 0.10373.$$

Then
$$\begin{aligned} t(\tau) &= \frac{c}{a} \sinh(0.10373) \approx \frac{2.99792458 \times 10^8}{9.8} \times 0.10379 \approx 3.204 \times 10^7; \text{ text{s}} \\ &= 1.016; \text{ text{yr}}. \end{aligned}$$

3. ****Distance.****
$$\begin{aligned} x &= \frac{c^2}{a} (\cosh(0.10373) - 1) \approx \frac{(2.99792458 \times 10^8)^2}{9.8} \times (1.00522 - 1) \approx 4.92 \times 10^{15}; \text{ text{m}} \\ &= 0.521; \text{ text{ly}}. \end{aligned}$$

4. ****Final speed.****
$$\beta = \tanh(0.10373) \approx 0.1032.$$

5. ****The student's error.****
The student's $\beta = a\tau/c \approx 0.10373 > 0.1032$ (correct). The mistake is that $\beta = \tanh(\eta)$, not η itself, so β saturates below 1.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{distance_ly} &= 0.521 \\ \mathrm{final_beta} &= 0.1032 \\ \mathrm{lab_time_yr} &= 1.016 \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.00000000 \ yr	FAIL	
distance_ly	0.563194	0.64362486 \ ly	FAIL	
final_beta	0.774547	0.76159416	FAIL	-0.0129528

Suggested error labels: distance_ly:UNPARSEABLE:parse-fail:TokenError;final_beta:MISMATCH;lab_time_yr:UNPARSEABLE:pars-fail:TokenError;unparseable_value

Failed fields: distance_ly;final_beta;lab_time_yr

Run ID: dc36a2b4-f64b-443a-b417-a32edd88612a

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation
1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER
$$\begin{aligned} \mathrm{lab_time_yr} &\approx 1.187045 \\ \mathrm{distance_ly} &\approx 0.563194 \\ \mathrm{final_beta} &\approx 0.774547 \end{aligned}$$
Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly ; β 0.77 ; rejects $v = a\tau$ — rapidity grows linearly, not velocity.
Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Exact hyperbolic motion.**** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the ****rapidity****.

2. ****Rapidity & lab time.**** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.00000000$. Lab time $= (c/a) \sinh(1.00000000) \Rightarrow$ ****1.00000000 yr**** (> 1 proper year — the correct dilation direction).

3. ****Distance.**** $x = (c^2/a)(\cosh(1.00000000) - 1) \Rightarrow$ ****0.64362486 ly****.

4. ****Final speed.**** $\beta = \tanh(1.00000000) =$ ****0.76159416****.

5. ****The student's error.**** $v = a\tau$ gives $\beta = a\tau/c = 1.00000000 > 1$ (superluminal). The resolution: the ****rapidity**** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

****FINAL ANSWER****

$$\begin{aligned} \mathrm{lab_time_yr} &\approx 1.00000000 \\ \mathrm{distance_ly} &\approx 0.64362486 \\ \mathrm{final_beta} &\approx 0.76159416 \end{aligned}$$

Question

A rocket maintains constant proper acceleration $a = 9.8 \text{ m/s}^2$ (as measured by an on-board accelerometer), starting from rest. Use $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$ and $1 \text{ ly} = c \times 1 \text{ yr}$.

Conventions: hyperbolic (Rindler) motion; τ is proper time on the rocket.

Tasks:
1. Write the exact expressions for lab time $t(\tau)$, lab distance $x(\tau)$, and velocity $\beta(\tau)$ for constant proper acceleration.
2. After $\tau = 1.00$ proper year, evaluate the elapsed lab time in years.
3. Evaluate the distance covered in light-years.
4. Evaluate the final speed $\beta = v/c$.
5. A student uses $v = a\tau$ and gets $\beta > 1$. Explain the error and why the correct β stays below 1.

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
lab_time_yr	1.187045	1.106	FAIL	-0.081045
distance_ly	0.563194	0.586	FAIL	0.022806
final_beta	0.774547	0.820	FAIL	0.045453

Suggested error labels:
distance_ly:MISMATCH;final_beta:MISMATCH;lab_time_yr:MISMATCH;numeric_mismatch

Failed fields: distance_ly;final_beta;lab_time_yr

Run ID: 8dd9d539-5113-456a-ad1b-2987c0bf85c9

Worked Gold Answer

SR_ROCKET_001 — Relativistic rocket: one proper year at 1 g

Given: proper acceleration $a = 9.8 \text{ m/s}^2$, from rest; $1 \text{ yr} = 3.15576 \times 10^7 \text{ s}$, $1 \text{ ly} = c \cdot \text{yr}$; hyperbolic (Rindler) motion, τ = proper time.

Derivation

1. **Exact hyperbolic motion.** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the **rapidity**.
2. **Rapidity & lab time.** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0316$. Lab time $= (c/a) \sinh(1.0316) \Rightarrow$ **1.187 yr** (> 1 proper year — the correct dilation direction).
3. **Distance.** $x = (c^2/a)(\cosh(1.0316) - 1) \Rightarrow$ **0.563 ly**.
4. **Final speed.** $\beta = \tanh(1.0316) =$ **0.7745**.
5. **The student's error.** $v = a\tau$ gives $\beta = a\tau/c = 1.03 > 1$ (superluminal). The resolution: the **rapidity** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

FINAL ANSWER

$$\begin{aligned} \backslash \mathrm{lab_time_yr} &= 1.187045 \\ \backslash \mathrm{distance_ly} &= 0.563194 \\ \backslash \mathrm{final_beta} &= 0.774547 \end{aligned}$$

Reasoning checkpoints: uses $x = (c^2/a)(\cosh-1)$, $t = (c/a)\sinh$, $\beta = \tanh$; $a\tau/c \approx 1.03$; lab time $1.19 > 1$ proper yr; distance 0.56 ly ; β 0.77 ; rejects $v = a\tau$ — rapidity grows linearly, not velocity.

Wrong-answer fingerprints: $v = a\tau / \beta > 1 \rightarrow$ 'C_NEWTONIAN_KINEMATICS_USED' / 'C_SUPERLUMINAL_CLAIM'; rapidity mishandled $\rightarrow$ 'N_RAPIDITY_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Exact hyperbolic motion.**** $t(\tau) = (c/a) \sinh(a\tau/c)$, $x(\tau) = (c^2/a)(\cosh(a\tau/c) - 1)$, $\beta(\tau) = \tanh(a\tau/c)$. The dimensionless combination $a\tau/c$ is the ****rapidity****.

2. ****Rapidity & lab time.**** $a\tau/c = 9.8 \cdot 3.15576 \times 10^7 / 2.99792458 \times 10^8 = 1.0516$. Lab time $= (c/a) \sinh(1.0516) \Rightarrow$ ****1.106 yr**** (> 1 proper year — the correct dilation direction).

3. ****Distance.**** $x = (c^2/a)(\cosh(1.0516) - 1) \Rightarrow$ ****0.586 ly****.

4. ****Final speed.**** $\beta = \tanh(1.0516) =$ ****0.820****.

5. ****The student's error.**** $v = a\tau$ gives $\beta = a\tau/c = 1.0516 > 1$ (superluminal). The resolution: the ****rapidity**** $a\tau/c$ grows linearly, but velocity $\beta = \tanh(\text{rapidity})$ saturates below 1. Newtonian kinematics wrongly adds velocities instead of rapidities.

****FINAL ANSWER****

$$\begin{aligned} \backslash \mathrm{distance_ly} &= 0.586 \\ \backslash \mathrm{final_beta} &= 0.820 \\ \backslash \mathrm{lab_time_yr} &= 1.106 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269	PASS	-0.000185
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: 863e22f3-01f3-468c-994c-0a868b72a3dc

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field overestimates T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Mean field.**** $k_B T_c^{\text{MF}}/J = z = 4$.

2. ****Onsager exact.**** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/\ln(2.414) = 2/0.881 = 2.269$.

3. ****Ratio.**** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = 0.567$. Mean field ****overestimates**** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. ****One dimension.**** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567 \\ \mathrm{tc_exact} &= 2.269 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269	PASS	-0.000185
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: a82bb41e-4662-4d67-95ff-851677b05abd

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

Derivation

1. **Mean field.** $k_B T_c^{\text{MF}}/J = z = 4$.

2. **Onsager exact.** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = 2.269$.

3. **Ratio.** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = 0.567$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. **One dimension.** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567 \\ \mathrm{tc_exact} &= 2.269 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269	PASS	-0.000185
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: 8032048a-8e04-432c-b3eb-aab06ea4d99e

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

Derivation

1. **Mean field.** $k_B T_c^{\text{MF}}/J = z = 4$.

2. **Onsager exact.** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = 2.269$.

3. **Ratio.** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = 0.567$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. **One dimension.** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567 \\ \mathrm{tc_exact} &= 2.269 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269	PASS	-0.000185
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: 5f22cbbd-f1ee-4af0-aa42-2c057e6b9037

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

Derivation

1. **Mean field.** $(k_B T_c^{\text{MF}})/J = z = 4$.

2. **Onsager exact.**
$$k_B T_c/J = \frac{2}{\ln(1+\sqrt{2})}$$

Numerically, $(\ln(1+\sqrt{2})) \approx 0.881373587$, so
$$k_B T_c/J \approx \frac{2}{0.881373587} \approx 2.269.$$

3. **Ratio.**
$$\frac{T_c^{\text{exact}}}{T_c^{\text{MF}}} = \frac{2.269}{4} \approx 0.567.$$

Mean field **overestimates** (T_c) because it replaces each spin's neighbors by their average, neglecting thermal fluctuations. In a low-dimensional system these fluctuations are strong enough to destroy long-range order at a lower temperature than the mean-field prediction.

4. **One dimension.**
The exact result is $(T_c = 0)$; there is no finite-temperature phase transition. Physically, a domain wall in 1D costs an energy of $(2J)$ but gains an entropic term $(-T \ln N)$ that dominates for any $(T>0)$ as the system size (N) grows, so thermal fluctuations always melt the ferromagnetic order.

FINAL ANSWER

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567 \\ \mathrm{tc_exact} &= 2.269 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269185314...	PASS	
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567353...	PASS	

Suggested error labels: none

Failed fields: none

Run ID: 7ccc748d-227f-4678-a6d4-a73a86a210d0

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Mean field.**** $k_B T_c^{\text{MF}}/J = z = 4$.

2. ****Onsager exact.**** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/\ln(1+\sqrt{2}) = 2/0.8813735870195429 = 2.269185314\dots$.

3. ****Ratio.**** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269185314\dots/4 = 0.567353\dots$. Mean field overestimates T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. ****One dimension.**** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567353\dots \\ \mathrm{tc_exact} &= 2.269185314\dots \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269,	PASS	-0.000185
tc_meanfield	4.0	4.000.	PASS	
ratio_exact_mf	0.567296	0.567,	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: 06aa69b6-039e-4f7c-91d6-191279776ccc

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact → ‘C_MEANFIELD_AS_EXACT’; claims a 1D finite-T transition → ‘C_1D_TRANSITION_CLAIMED’; wrong Onsager value → ‘N_ONSAGER_VALUE_ERROR’.

Full Raw Model Response

****Derivation****

1. ****Mean field.**** $(k_B T_c^{\text{MF}})/J = z = \boxed{4}$.

2. ****Onsager exact.**** $(k_B T_c/J = 2/\ln(1+\sqrt{2})) = 2/\ln(1+1.4142) = \boxed{2/\ln(2.4142)} \approx \boxed{2.269}$.

3. ****Ratio.**** $(T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \boxed{0.567})$. Mean field **overestimates** (T_c) because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. ****One dimension.**** $(T_c = 0)$ – no finite-temperature transition. A single domain wall costs energy $(2J)$ but gains entropy $(k \ln N)$; for any $(T > 0)$ the free-energy term $(-T\Delta S)$ wins for large (N) , so order is always destroyed (Peierls / domain-wall argument).

****FINAL ANSWER****

$$\begin{aligned} & \backslash \begin{aligned} & \mathrm{ratio_exact_mf} \quad \&= 0.567, \backslash \backslash \\ & \mathrm{tc_exact} \quad \&= 2.269, \backslash \backslash \\ & \mathrm{tc_meanfield} \quad \&= 4.000. \backslash \backslash \end{aligned} \\ & \backslash \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	3.6434	FAIL	1.37422
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.9109	FAIL	0.343604

Suggested error labels:
ratio_exact_mf:MISMATCH;tc_exact:MISMATCH; numeric_mismatch

Failed fields: ratio_exact_mf,tc_exact

Run ID: 1578b708-15cc-4472-862d-060fbd5e95c

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation
1. **Mean field.** $k_B T_c^{\text{MF}}/J = z = 4$.
2. **Onsager exact.** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.
3. **Ratio.** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.
4. **One dimension.** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER
$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Mean field.**** $k_B T_c^{\text{MF}}/J = z = \mathbf{**4**}$.

2. ****Onsager exact.**** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/\ln(1+1.4142) \approx 2/0.5493 = \mathbf{**3.6434**}$.

3. ****Ratio.**** $T_c^{\text{exact}}/T_c^{\text{MF}} = 3.6434/4 = \mathbf{**0.9109**}$. Mean field ****overestimates**** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. ****One dimension.**** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.9109 \\ \mathrm{tc_exact} &= 3.6434 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$

Question

Consider the ferromagnetic Ising model on the square lattice (coordination number $z = 4$), coupling $J > 0$, zero field.

Conventions: report critical temperatures as the dimensionless ratio $k_B T_c/J$.

Tasks:
1. Write down the mean-field estimate $k_B T_c^{\text{MF}}/J = z$ and evaluate it.
2. State Onsager's exact result $k_B T_c/J = 2/\ln(1+\sqrt{2})$ and evaluate it.
3. Compute the ratio $T_c^{\text{exact}}/T_c^{\text{MF}}$. Why does mean field overestimate T_c ?
4. What happens in one dimension, and why? (State the exact 1D result and the physical reason.)

Numeric Comparison

field_name	expected	model_value	match	delta_or_status
tc_exact	2.269185	2.269	PASS	-0.000185
tc_meanfield	4.0	4	PASS	0
ratio_exact_mf	0.567296	0.567	PASS	-0.000296

Suggested error labels: none

Failed fields: none

Run ID: 9e6a063d-dc82-49ce-aceb-7ac5aad0e02b

Worked Gold Answer

STAT_ISING_001 — 2D Ising: Onsager’s exact T_c vs mean field

Given: ferromagnetic Ising on the square lattice (coordination $z = 4$), $J > 0$, zero field. Report $k_B T_c/J$.

Derivation

1. Mean field. $k_B T_c^{\text{MF}}/J = z = 4$.

2. Onsager exact. $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.88137 = \mathbf{2.269}$.

3. Ratio. $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = \mathbf{0.567}$. Mean field **overestimates** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. One dimension. $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

FINAL ANSWER

$$\begin{aligned} \mathrm{tc_exact} &= 2.269185 \\ \mathrm{tc_meanfield} &= 4.0 \\ \mathrm{ratio_exact_mf} &= 0.567296 \end{aligned}$$

Reasoning checkpoints: MF $k_B T_c/J = z = 4$; Onsager $2/\ln(1+\sqrt{2}) = 2.269$; ratio ≈ 0.567 (MF neglects fluctuations); 1D $T_c = 0$ (domain-wall entropy); does not present the mean-field value as exact.

Wrong-answer fingerprints: presents mean-field as exact $\rightarrow$ 'C_MEANFIELD_AS_EXACT'; claims a 1D finite-T transition $\rightarrow$ 'C_1D_TRANSITION_CLAIMED'; wrong Onsager value $\rightarrow$ 'N_ONSAGER_VALUE_ERROR'.

Full Raw Model Response

****Derivation****

1. ****Mean field.**** $k_B T_c^{\text{MF}}/J = z = 4$.

2. ****Onsager exact.**** $k_B T_c/J = 2/\ln(1+\sqrt{2}) = 2/0.8814 = 2.269$.

3. ****Ratio.**** $T_c^{\text{exact}}/T_c^{\text{MF}} = 2.269/4 = 0.567$. Mean field ****overestimates**** T_c because it replaces each spin's neighbors by their average and ignores fluctuations; thermal fluctuations destroy long-range order at a lower temperature than the mean-field estimate.

4. ****One dimension.**** $T_c = 0$ — no finite-temperature transition. A single domain wall costs energy $2J$ but gains entropy $k \ln N$; for any $T > 0$ the free-energy term $-T\Delta S$ wins for large N , so order is always destroyed (Peierls / domain-wall argument).

****FINAL ANSWER****

$$\begin{aligned} \mathrm{ratio_exact_mf} &= 0.567 \\ \mathrm{tc_exact} &= 2.269 \\ \mathrm{tc_meanfield} &= 4 \end{aligned}$$